



Project Report On

Veridian - An AI powered Financial Advisor

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CERTIFICATE

*This is to certify that the project report entitled "**Veridian - An AI powered Financial Advisor**" is a bonafide record of the work done by **Navaneeth K Bijumon (U2203150)**, **Neha George (U2203152)**, **Sarath Menon Girish (U2203190)** and **Vijai Kiren S (U2203209)**, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

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Abstract

This paper presents Veridian, an AI-driven financial advisory system designed to support intelligent portfolio management through interpretable Buy, Hold, and Sell recommendations. The system adopts a teacher–student ensemble architecture in which multiple specialized analytical models independently evaluate a stock using diverse financial perspectives, including technical analysis, fundamental analysis, sentiment analysis, alternative web data, macroeconomic indicators, market cycle assessment, and expert consensus signals.

Each student model produces probabilistic predictions along with implicit confidence measures. These predictions are then aggregated by a central teacher model using a reliability-weighted decision mechanism to generate stable and robust investment recommendations.

Veridian integrates automated multi-source data collection, preprocessing pipelines, and parallel inference, along with an explainable reporting layer that converts model outputs into human-readable insights delivered through a conversational chatbot interface.

Experimental evaluation on real-world stocks demonstrates that the proposed architecture improves decision stability, effectively reconciles conflicting analytical signals, and enhances interpretability compared to single-model approaches. By combining heterogeneous financial intelligence with explainable AI, Veridian bridges the gap between complex investment analytics and accessible retail decision support, providing a scalable and user-centric intelligent advisory framework.

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List of Abbreviations

AI	- Artificial Intelligence
API	- Application Programming Interface
CAB-LSTM	- Context-Aware Bidirectional Long Short-Term Memory
CNN	- Convolutional Neural Network
EPS	- Earnings Per Share
ESG	- Environmental, Social, and Governance
FRED	- Federal Reserve Economic Data
GDP	- Gross Domestic Product
JSON	- JavaScript Object Notation
LSTM	- Long Short-Term Memory
MAE	- Mean Absolute Error
MACD	- Moving Average Convergence Divergence
ML	- Machine Learning
NLP	- Natural Language Processing
OHLCV	- Open, High, Low, Close, Volume
P/E	- Price-to-Earnings Ratio
RAM	- Random Access Memory
RMSE	- Root Mean Square Error
ROE	- Return on Equity
RSS	- Really Simple Syndication
SHAP	- SHapley Additive exPlanations
SMA	- Simple Moving Average
TCN	- Temporal Convolutional Network
TFT	- Temporal Fusion Transformer
VIX	- Volatility Index
XGBoost	- Extreme Gradient Boosting

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Chapter 1

Introduction

Artificial Intelligence has greatly changed the landscape of the Financial Technology sector by facilitating the development of intelligent systems that can process vast amounts of financial data and help investors make informed decisions. Today, investors depend on various sources such as market indicators, expert views, financial reports, and news analysis for decision-making regarding the performance of stocks in the market. However, processing such varied sources of information can be challenging and often involves human biases and complexities.

Veridian has been developed as an AI-based investment assistant that can offer intelligent support for stock analysis and recommendation through a conversational interface. The system has been integrated with various machine learning algorithms and techniques for financial data analysis and natural language processing that can offer Buy, Sell, or Hold recommendations for stocks. The system has been integrated with various perspectives for analysis, including technical analysis, fundamental analysis, sentiment analysis, and market risks for more accurate predictions.

1.1 Background

The latest advancements in the field of machine learning, ensemble learning methods, and natural language processing have made it possible to build intelligent financial analysis systems capable of processing large-scale financial market data to obtain meaningful insights. Modern financial analytics systems use predictive models and automated analysis to assist investors in making investment decisions by evaluating the market opportunities.

The existing stock recommendation systems use a single analytical perspective, which may result in biased predictions. To improve the predictions of the system, ensemble learning methods can be applied. In the ensemble learning method, the system uses

multiple models to predict the financial market. Each of the models processes different aspects of the financial market. The predictions of the models are then integrated by another model to obtain the final prediction.

Veridian follows the same method by using multiple student models to process the financial market data. The student models process the financial market data using different aspects of the financial market, such as technical indicators, market sentiment, financial fundamentals, analyst recommendations, time series, and market risk. The predictions of the student models are then integrated by the teacher model, which is a Random Forest model. The teacher model provides the final investment recommendation along with a confidence score.

1.2 Problem Definition

Investors usually face challenges in processing vast amounts of financial data when making investment decisions. In evaluating stocks, various types of data must be interpreted, such as historical price trends, financial reports, analysts' views, and news sentiment. This is a challenging task for ordinary investors because it demands specialized knowledge and time to thoroughly analyze financial reports. In this context, the objective of this project is to build a tool that utilizes AI technology to help investors make investment decisions through automated stock analysis. The tool will provide recommendations to buy, sell, or hold stocks.

1.3 Scope and Motivation

The scope of the Veridian system includes developing an intelligent stock analysis platform based on artificial intelligence, which utilizes various machine learning models to assess financial data available in the financial market. The system analyzes various types of stock-related data available on the internet using specialized models.

The motivation behind developing the Veridian intelligent stock analysis system is to facilitate investment decisions by developing an intelligent assistant that can automatically analyze complex financial data. The intelligent assistant is based on an ensemble learning model that combines various analytical techniques to provide a balanced investment decision. Additionally, it also aims to increase transparency.

Table 1.1: Core Components of the Veridian System

Component	Technology Used	Purpose
Frontend	React.js	User interface for chatbot interaction and stock query visualization
Backend	Python / FastAPI	Handles API requests, data processing, and integration of prediction models
Student Models	Machine Learning Models	Perform specialized financial analysis such as technical indicators, sentiment analysis, and fundamental analysis
Teacher Model	Random Forest	Aggregates predictions from multiple student models to generate the final recommendation
Database	Firebase / Cloud Storage	Stores user queries, historical predictions, and system data

1.4 Objectives

1. To design and develop an AI-powered investment assistant capable of analyzing stock market data.
2. To implement multiple student models that evaluate different aspects of stock performance.
3. To develop a Random Forest-based teacher model that aggregates predictions from student models.
4. To generate Buy, Sell, or Hold recommendations along with confidence scores.
5. To build an interactive chatbot interface for user queries and financial insights.
6. To provide explainable reasoning for each investment recommendation.

1.5 Challenges

The process of developing an AI-based financial recommendation system has several challenges. The first challenge is to effectively combine various data sources such as historical price data, financial data, analyst ratings, and text-based sentiment data. The second challenge is to develop an efficient ensemble model that combines predictions made by various student models without any bias or overfitting.

It is also essential to ensure that the recommendation made by the financial recommendation system is transparent. Efficiently processing financial data in real-time and maintaining system scalability are also important considerations during system development.

1.6 Assumptions

The following assumptions were considered during the development of the Veridian system:

- Users access the system through a web-based interface.
- Financial data sources used by the system provide reliable and up-to-date information.
- Users have basic knowledge of stock market concepts such as Buy, Sell, and Hold recommendations.
- The system provides decision support and does not replace professional financial advice.

1.7 Societal / Industrial Relevance

Veridian offers an example of how artificial intelligence can be used in financial technology to support investors in analyzing various market data. The system enables the user to understand financial information more easily by automating financial analysis and utilizing various predictions.

From an industrial perspective, AI-powered financial assistants like Veridian can be integrated into brokerage platforms, financial advisory tools, and investment management systems. Such platforms can improve investment accessibility, support data-driven decision-making, and enhance transparency in financial recommendations.

1.8 Organization of the Report

This report is organized into six chapters. Chapter 1 introduces the background, objectives, and motivation behind the Veridian system. Chapter 2 presents a literature survey of existing AI-based financial analysis and recommendation systems. Chapter 3 describes the system architecture, model design, and technologies used in the implementation. Chapter 4 discusses the implementation details of the student models, teacher model, and system integration. Chapter 5 presents the results and evaluation of the developed system. Finally, Chapter 6 concludes the report and outlines possible future improvements.

1.9 Summary of Chapter

This chapter introduced Veridian, an investment assistant powered by AI technology with the capability to perform intelligent stock analysis and recommendation support, was introduced. The chapter has also covered the background of AI-driven financial analytics systems, the problem statement of the project, the objectives of the project, the scope of the project, the challenges involved, and the overall report structure.

Chapter 2

Literature Survey

This chapter provides an overview of the latest trends in the field of financial advisory systems using AI techniques, with a specific emphasis on the use of macroeconomic variables, technical/fundamental analysis, sentiment analysis of financial news articles and social media, alternative web data, financial market cycle identification, and risk-aware portfolio signals. The main aim of the current chapter is to provide an overview of the role of the aforementioned techniques in the development of sound financial decision models. The current chapter has also drawn inspiration from other benchmark articles like “Macroeconomic Forecasting and Machine Learning,” “Hybrid Deep Learning Models for Financial Time Series Forecasting,” “Stock Market Prediction Based on Financial News, Text Data Mining, and Investor Sentiment Analysis,” “Harnessing Alternative and Web Data for Financial Forecasting,” among others. The inspiration for the current chapter has motivated the development of the latest version of the Veridian financial advisor system, which incorporates an expanded ensemble of six analytical models to provide the teacher model with the capability to synthesize the results obtained by the analytical models to provide sound financial decision models.

2.1 Review of Key Research Works

2.1.1 Macroeconomic Model

The importance of **machine learning** in handling huge volumes of high-dimensional data, complicated models, and non-linear economic risks was, according to the authors of “Macroeconomic Forecasting and Machine Learning,” underscored by the recent developments in macroeconomic forecasting. Econometric models fit for the purpose of carrying out macroeconomic forecasts cannot handle numerous predictors and non-linear models, as these models are limited to linear models of time series; nevertheless, the authors pro-

posed a hybrid approach in accordance with the following three principles: regularization, out-of-sample validation, and the availability of flexible learning architectures to enhance the precision and consistency of economic risks forecasts.

Input and Datasets: The authors used a variety of high-dimensional macroeconomic estimators consisting of 130 predictors, such as labor market predictors like the number of jobs created, industrial production, aggregate price indices, credit, as well as other financial indicators, to build an empirical model for the prediction of the US economy. This type of large-dimensional dataset is available on FRED on an annual basis. This dataset is a standard reference for empirical macroeconomics.

Methodology: The authors use deep neural networks with quantile regression as the basis for the determination of the conditional probabilities of the macroeconomic variables. The authors' method also includes other activities that are important for the determination of the same. The activities include recursively determining the parameters of the models, employing cross-validation to determine the hyperparameters, as well as employing the shrinkage penalties to avoid overfitting. The use of nonlinear activation functions, for example, Leaky ReLU, results in small gains compared to the performance of the highly regularized linear models. The use of regularization with the help of the penalty mechanism results in large gains.

Findings: The findings reveal that the models using ML techniques are able to identify emerging downside risks in real time, such as changes in unemployment risk profiles. The improvement in forecasting is seen to be highest in the upper quantiles of the distribution where the predictor variables are stronger. The out-of-sample validation is also used to ensure stability in the models, and the optimal model complexity is used to balance variance reduction and forecasting accuracy.

Conclusion: The ML techniques using techniques such as shrinkage, distributional forecasting, and validation provide a framework to anticipate macroeconomic risks and thereby strengthen investment and policy decisions by not only looking at the mean forecasts but also the distribution of risks.

2.1.2 Technical and Company Fundamentals Model

This model employs the capabilities of deep temporal sequence learning and interpretable ensemble techniques for the evaluation and prediction of stock performance. The pro-

posed model is based on the development of hybrid models that combine temporal pattern recognition and structured financial indicators for the evaluation and prediction of stock performance. The proposed model employs the capabilities of the **LSTM** temporal encoder and the **LightGBM** decision layer for the evaluation and prediction of stock performance. The conceptual background for the proposed model can be based on the development of hybrid deep learning and gradient boosting for the evaluation and prediction of financial performance, as discussed in “Hybrid Deep Learning Models for Financial Time Series Forecasting” [2].

Input and Dataset: The datasets comprise daily and hourly **OHLCV** financial data of the **S&P 500** constituents, along with quarterly company fundamentals like **EPS**, **PE Ratio**, **ROE**, revenue growth, debt to equity ratio, etc. In addition to the above, sector-level features like market capitalization, median sector ratios, analyst estimate revisions have also been included to provide a holistic view of the firm-level data. The data was collected through various **APIs** like **Yahoo Finance**, **Tiingo**, **Alpha Vantage**, ensuring timely availability of the data. Missing values were handled by using the time forward method, while categorical data was one-hot encoded to incorporate it into the model.

Methodology: The proposed hybrid pipeline combines sequential and tabular learning through a two-stage process. In the first stage, the **LSTM (Long Short-Term Memory)** encoder processes temporal sequences of price-based inputs, capturing short- and long-term dependencies in financial time series. The encoder outputs latent embeddings that summarize dynamic relationships in historical market data.

In the second stage, these embeddings are concatenated with company fundamental features and fed into a **LightGBM** ensemble learner. LightGBM’s gradient boosting framework efficiently models non-linear feature interactions, handles missing values and categorical features, and maintains interpretability through **SHAP**-based feature attribution. The ensemble outputs **N-day forward returns** and corresponding confidence scores, which are translated into actionable **Buy, Hold, or Sell** recommendations based on dynamic thresholds and volatility-adjusted risk metrics. Data preprocessing includes z-score normalization, volatility scaling, and rolling-window feature generation. A walk-forward cross-validation approach was used to preserve temporal integrity and mitigate lookahead bias.

Findings: Experimental evaluation showed that the **LSTM + LightGBM** hybrid

outperformed individual models. Compared to **Pure LightGBM** and other temporal models such as **TCN**, **N-BEATS**, and **TFT**, the **LSTM + LightGBM** hybrid was able to attain an average test accuracy of 74.5% and an F1 score of 75.1 % on **EWZ** and **SPY** daily series, which are highly volatile stock time series. The addition of fundamental variables made the model more robust during regime changes, and the **LSTM** component was able to pick up on short-term patterns that other models might have otherwise ignored. Evaluation criteria such as **MAE**, **RMSE**, and directional hit ratio showed that the model was able to perform consistently and was statistically significant. Using the **SHAP** package, we also showed that short-term momentum indicators such as EPS growth

Conclusion: The Technical and Company Fundamentals hybrid model offers a solid analytical background for **AI-based portfolio advisory systems** in the future. By using the temporal extraction capabilities of the LSTM and the interpretability and stability of the LightGBM algorithm, the model can provide decision-making signals that take into account both the short-term market trends and the intrinsic value of the companies in the portfolio for platforms such as **Veridian** and other portfolio advisory systems in the future. The incorporation of macroeconomic and sentiment embeddings can be part of the future work for the improvement of the model.

2.1.3 News and Sentiment Analysis Model

The News and Sentiment Analysis Model aims at understanding the psychology of investors and the market through the analysis of textual data available in the form of news and social media platforms. The study "Stock Market Prediction Based on Financial News, Text Data Mining, and Investor Sentiment Analysis" [3] has suggested the Context-Aware Bidirectional LSTM (CAB-LSTM), which can effectively analyze the market through the incorporation of attention mechanisms and bidirectional temporal learning.

Input and Dataset: The datasets are collected from multiple news and media sources, including Reuters, Bloomberg, and CNBC, as well as social platforms such as Twitter and StockTwits. Each text entry is labeled according to sentiment polarity (positive, neutral, negative) or market impact categories. Data preprocessing includes deduplication and timestamp alignment with market movements.

Methodology: Preprocessing of text data includes tokenization, stop word removal, and lemmatization. Word embedding techniques, such as Word2Vec or financial embedding models like FinBERT, are used for contextual vectorization. The proposed CAB-LSTM model is an extension of the traditional BiLSTM model, which incorporates a context attention layer to assign importance weights to relevant terms or entities, thus improving the accuracy of sentiment extraction. The proposed model can focus on both local context (i.e., phrase) and global context (i.e., tone of the news).

Findings: The proposed CAB-LSTM model has better performance than the baseline LSTM model and CNN model in sentiment classification and market movement prediction tasks. Experiments have shown that the proposed model, by incorporating sentiment scores with quantitative features, can increase the accuracy of the model by around 12%. Furthermore, the context attention mechanism has the advantage of providing interpretability by identifying the key keywords that affect the sentiment triggers in the articles.

Conclusion: This model enhances the capacity of Veridian to incorporate qualitative market intelligence to support adaptive decision-making with real-time investor sentiment and media tone. The interpretability of attention is also consistent with the emphasis of Veridian on transparent decision recommendations.

2.1.4 Alternative and Web Data Model

The Alternative and Web Data Model utilizes alternative data types such as search engine trends, ESG scores, satellite imagery, and web traffic data to gain market insights. The importance of alternative data types is emphasized in a study titled “Harnessing Alternative and Web Data for Financial Forecasting” [4]. The study explains how alternative data types are gaining importance to improve forecasting accuracy when traditional financial data is limited or not available.

Input and Dataset: The model ingests multiple alternative data sources, including Google Trends (search volumes for specific tickers and sectors), Wikipedia page views (investor curiosity metrics), ESG scores (sustainability and governance performance), e-commerce sales indicators, and job posting frequencies (labor market strength). These inputs capture behavioral, environmental, and sentiment-based cues that often precede observable market movements.

Methodology: The workflow process starts with the feature engineering and extraction process, which includes natural language processing for text-based data, correlation filtering for feature selection, and anomaly detection for unusual activity spikes. The predictive process utilizes machine learning models like XGBoost, Random Forest, and clustering to detect underlying patterns and sectoral relationships. To improve the predictability of the models, data fusion techniques are used to combine alternative data with traditional macroeconomic and technical data to create a unified feature space.

Findings: As verified through empirical testing, incorporating alternative data can help in improving early detection of inflection points and changes in markets and sectors. For example, sudden spikes in Google searches or deviations in ESG scores have shown consistent correlation with changes in stock momentum and volatility. The fusion of structured and alternative features using ensemble algorithms has shown improved precision in forecasting and risk-adjusted returns.

Conclusion: The Alternative and Web Data Model adds a novel intelligence layer to the Veridian analytics ecosystem by fusing unstructured behavioral data with structured financial indicators. This integration enables more adaptive and data-driven forecasting, particularly under rapidly changing market conditions.

2.2 Comparative Analysis

The reviewed studies collectively illustrate that integrating diverse data modalities—macroeconomic, technical and fundamental, sentiment, and alternative web data—substantially improves financial forecasting accuracy. Table 2.1 summarizes their methodological differences, data sources, and key outcomes.

Model / Paper	Focus Area	Data Sources	ML / Analytical Methods	Key Findings
Macroeconomic Model	Forecasting economic trends and systemic risk	GDP, Inflation, Interest Rates, Unemployment, Government Debt, Trade Balance	LSTM	Improved forecasting accuracy (10–15%) and interpretable macroeconomic signals
Technical & Fundamental Model	Stock performance using price and financial indicators	OHLCV data, RSI, MACD, EPS, P/E, ROE, Revenue Growth	LightGBM + LSTM Hybrid	Outperformed standalone models with 74.5% accuracy and robust predictions during regime shifts; SHAP analysis showed EPS and sector ratios as key drivers
News & Sentiment Model	Market movement prediction using text data	Financial news, social media posts (Twitter, StockTwits)	CAB-LSTM with FinBERT embeddings	Sentiment integration improves accuracy by 12%; attention mechanism enhances interpretability by highlighting influential terms
Alternative & Web Data Model	Market forecasting using non-traditional data	Google Trends, ESG scores, Web traffic, Patent filings, Job postings	ALTFusion + XGBoost	Early detection of market inflection points; improves forecasting precision by fusing structured and behavioral data

Table 2.1: Comparative summary of AI-based models for financial market prediction in Veridian

2.3 Summary and Research Gaps

The literature reviewed demonstrates the immense potential of machine learning in multi-source financial forecasting. By combining macroeconomic, technical, sentiment, and alternative data, models can capture both quantitative and qualitative market drivers, leading to improved predictive performance and actionable investment strategies.

However, several research and implementation gaps remain:

- **Data Integration Challenges:** Combining heterogeneous data sources (structured, unstructured, and alternative) remains computationally intensive and complex.
- **Interpretability:** Many high-performing models, especially deep neural networks, lack transparency in explaining predictions to investors.
- **Real-Time Adaptability:** Most models rely on historical datasets; adapting them to fast-changing financial environments remains a challenge.
- **Generalization and Bias:** Data-driven models may overfit to specific markets or time periods, requiring continuous retraining and validation.

Addressing these gaps will enable the Veridian system to evolve into a robust, interpretable, and adaptive financial advisor capable of providing intelligent, data-informed investment recommendations.

Chapter 3

Proposed Method

This chapter describes the methodology used to design and implement the Veridian financial advisory system. The proposed system will employ a teacher-student ensemble model coupled with natural language processing and multiple financial data sources to create intelligent investment recommendations. The proposed method will integrate several analytical models that will work concurrently to evaluate the markets from different perspectives. The results obtained will then be combined through a reliability-weighted decision model to arrive at a recommendation, such as Buy, Hold, or Sell.

3.1 System Workflow Overview

The overall workflow of the Veridian system starts when a user poses an investment-related query through the conversational interface. The system processes the query to identify the company for which the query is made and retrieves the relevant financial data associated with the company.

Once the stock is identified, the system starts an automated pipeline to gather and process financial data from multiple sources related to the stock. This financial data is then used as input to a series of specialized models called student models. Each student model analyzes the stock from a different perspective and generates a probabilistic prediction regarding the investment action.

The predictions made by the student models are aggregated using a reliability-weighted ensemble strategy by a specialized model called the teacher model. The teacher model generates a probability distribution over the three actions: Buy, Hold, and Sell. The action with the highest probability is selected as the final recommendation.

3.2 NLP-Based Query Understanding and Stock Resolution

The Veridian system provides users with the ability to interact with the system through a query based on a natural language query. When the user provides a query, the system processes the query to identify important entities such as the company, the stock symbol, or the investment intent through the use of natural language processing techniques.

The identified entity is then converted to a standard symbol representing a publicly traded stock on a financial exchange. This provides consistency to the data retrieval mechanism to avoid ambiguity in case the companies have similar names.

After the stock symbol has been resolved, the system sends the ticker symbol to the analytical pipeline to start the retrieval of the financial data.

3.2.1 Automated Data Collection and Preprocessing

Once the target stock is identified, the system triggers its data acquisition mechanism. The system retrieves data from various sources, which include historical stock prices, trading volumes, financial statements, economic indicators, and articles and social media posts.

The acquired data is subject to a series of preprocessing steps to validate its correctness and consistency. These preprocessing techniques include data normalization, removal of missing values, reduction of noise, and synchronization of data with different sampling rates.

Technical indicators such as moving averages and volatilities are computed using historical price data. Text data sources such as articles and social media posts are tokenized and filtered to remove stop words before subjecting them to sentiment analysis.

3.3 Student Model Architecture and Parallel Inference

The analytical heart of the Veridian system is comprised of various student models that run concurrently. Each student model analyzes the stock based on a particular financial perspective, including fundamental analysis, sentiment analysis, macro environment, alternative data signals, market cycles, and consensus of expert analysts.

Each student model independently derives a probability distribution that reflects the probability of the stock being classified as a buy, a hold, or a sell. The probabilities also

reflect the analytical confidence of each model. The concurrent operation of various student models makes the process computationally efficient, enabling the system to process various financial perspectives.

3.4 Teacher Model and Decision Aggregation

The teacher model serves as the central decision-making component of the Veridian system. It aggregates the predictions produced by all student models using a reliability-weighted ensemble approach.

Each student model produces a probability vector representing Buy, Hold, and Sell outcomes. The teacher model assigns predefined reliability weights to each analytical dimension based on the robustness and reliability of the underlying model.

The final aggregated probability distribution is computed by combining the weighted outputs of all student models. The final investment recommendation is determined by selecting the action with the highest aggregated probability. This approach ensures that stronger analytical signals have greater influence on the final decision.

3.5 Explanation Generation and Result Reporting

To improve the transparency of the system and build trust with the users, the Veridian system has an explanation generation component. This component converts the numerical predictions made by the analytical models into investment insights understandable by the user.

The system provides the user with an overview of the key factors on which the investment decision is based, such as the analytical signals of the student models. The explanation component of the system provides the user with a report on the basis of the Buy, Hold, or Sell investment decision.

The final investment recommendation, confidence score, and explanation report are presented to the user through the chatbot interface, allowing investors to understand the rationale behind each decision.

3.6 Summary of the Chapter

This chapter presented the proposed methodology used in the Veridian financial advisory system. The system workflow begins with natural language query processing and stock identification, followed by automated data collection and preprocessing from multiple financial data sources. The processed data is analyzed using several specialized student models that evaluate different financial indicators such as sentiment, macroeconomic signals, alternative data, and market trends.

The predictions made by the models are then aggregated using a teacher-student ensemble architecture to produce a final investment recommendation. The chapter also introduced the mechanism of explanation generation to produce investment insights from the analysis results. The proposed methodology is to be used by Veridian to integrate various financial signals and produce a stable and data-driven investment recommendation.

Chapter 4

System Design

4.1 Overall Architecture

The architecture of the Veridian system adopts a modular structure that allows for the implementation of intelligent analysis of finances and the processing of data in an expansive manner. The system comprises several modules that combine the capabilities of the web interface, the backend services, the machine learning models, and the data management modules in order to produce stock market recommendations.

The architecture of the system comprises three main layers: the presentation layer, the application layer, and the data processing layer. The presentation layer offers the user an interface that allows for the interaction of the user and the system through which the user can ask stock-related questions and obtain the analysis results. The application layer offers the functionality for API requests and the processing of the data using the models. The data processing layer comprises several student models that analyze different types of financial data and the teacher model that integrates the results in order to produce the final recommendation.

The ensemble architecture offers the benefits of increasing the reliability of the predictions through the combination of different perspectives in the analysis process, including the technical analysis, the market sentiment analysis, and the financial analysis.

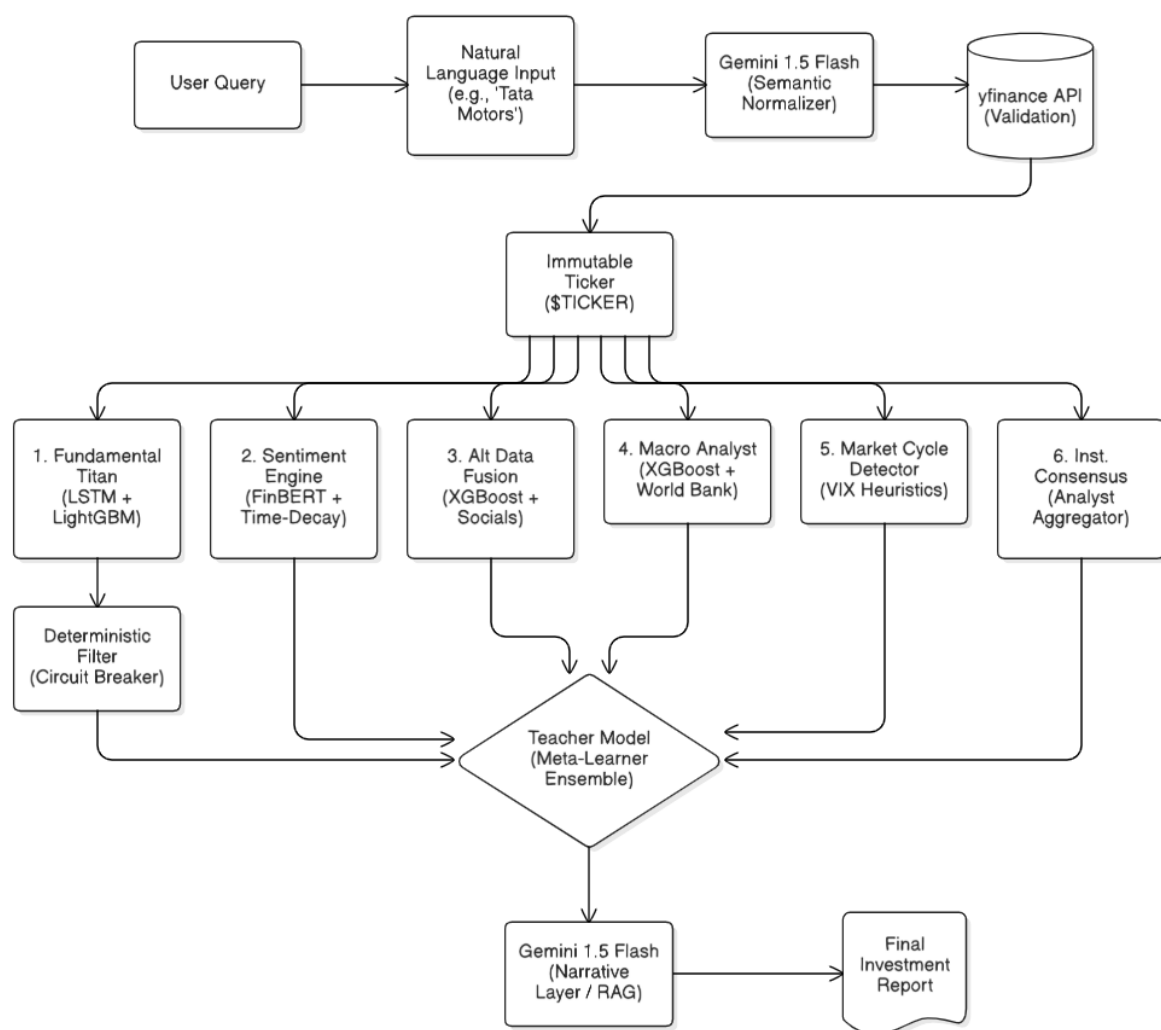


Figure 4.1: Teacher–Student Architecture of the Veridian Financial Advisory System

4.2 Component Design

The Veridian system is composed of several interconnected components that work together to provide intelligent stock analysis.

- **User Interface:** The frontend provides a user-friendly dashboard where investors can search for stocks, view predictions, and access explanations for recommendations.
- **API Layer:** The backend server handles communication between the user interface and the machine learning models. It processes user requests, retrieves financial data, and triggers the prediction pipeline.

- **Prediction Engine:** This module runs multiple student models that analyze stock market data using different analytical approaches.
- **Aggregation Module:** A teacher model combines predictions from student models and generates a final Buy, Sell, or Hold recommendation.

4.2.1 Prediction Model Integration

The prediction engine of the Veridian system is based on a teacher-student ensemble architecture that uses multiple specialized analytical models to run in parallel to assess different aspects of market behavior. The specialized models used in the prediction engine are referred to as student models and are designed to individually analyze financial data and produce probabilistic predictions regarding the classification of a stock as **Buy**, **Hold**, and **Sell**.

Each student model is designed to produce a probability vector indicating the likelihood of the three possible investment actions. The degree of concentration of the probability distribution is also a measure of the confidence level of the analytical model used to produce the prediction. The use of a unified probabilistic model enables the results of multiple analytical models to be aggregated in a uniform manner.

The Veridian system uses multiple specialized student models to cover different aspects of financial intelligence and include models for performing technical and fundamental analysis, sentiment analysis of financial news and social media posts, alternative and web-based behavioral signals, macroeconomic analysis, market cycle analysis, and consensus analysis from financial analyst reports.

After all the student models finish their analysis, a decision aggregation mechanism based on a reliability-weighted decision aggregation scheme is used to aggregate the probabilistic results of all student models by the teacher model. The teacher model then combines the probability distributions output by the student models using a set of pre-determined weights on each analytical dimension. The final investment decision is made based on the action with the maximum probability.

The proposed framework for making predictions based on an ensemble of models allows Veridian to leverage heterogeneous financial data sources while maintaining stability in decision making.

4.3 Design Diagrams

4.3.1 Sequence Diagram

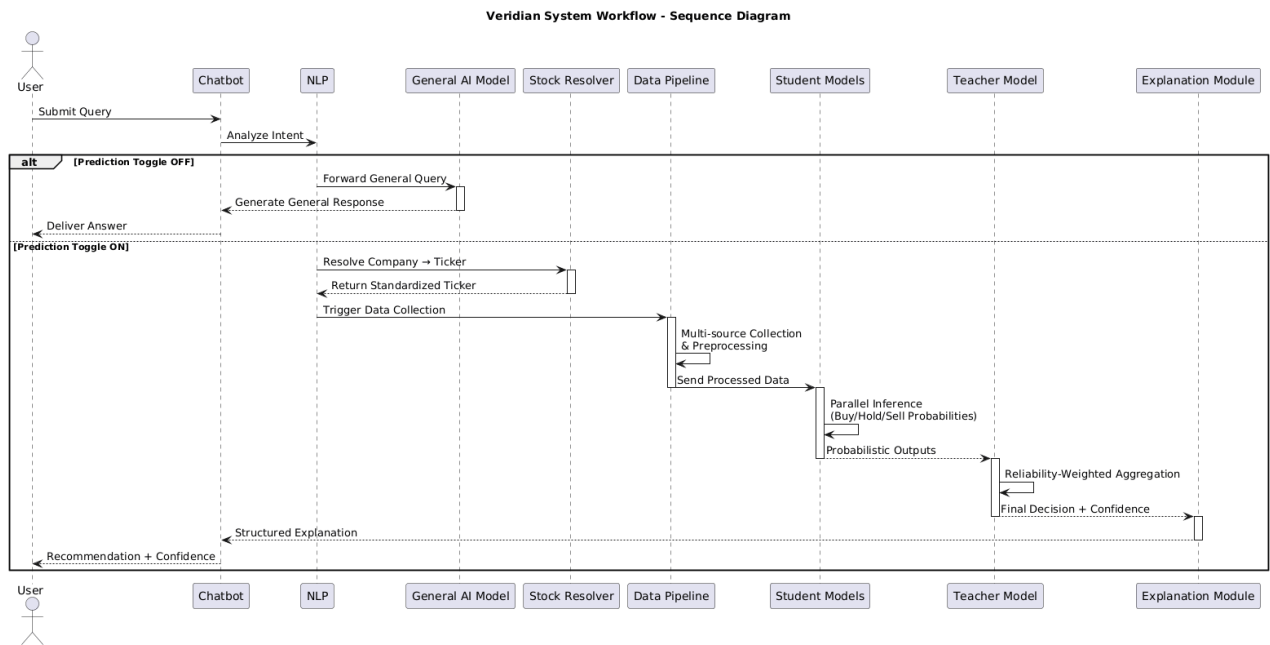


Figure 4.2: Sequence Diagram

The sequence diagram provided in Figure 4.2 describes the workflow of the Veridian financial advisory system when a user submits a query pertaining to a stock through the chatbot interface.

The query is first passed through the Natural Language Processing module, where the prediction feature will identify the user’s intention and extract the company name or ticker symbol associated with the query.

If the prediction feature is enabled, the stock resolver will standardize the company name into a standard ticker symbol. Next, the data pipeline will gather the financial data from multiple sources, where the preprocessing will take place. This will allow the multiple student models to access the processed data, where the probability of a Buy, Hold, or Sell signal will be generated.

The teacher model will then integrate the predictions through a reliability-weighted aggregation, where the final investment recommendation will be provided, along with the confidence score. Finally, the explanation module will provide a structured explanation, where the chatbot will communicate the results to the user.

4.4 Module Division

The Veridian system is implemented using a modular architecture in which different analytical models evaluate various dimensions of financial market behavior. Each analytical component is implemented as a **student model** that independently evaluates the stock and generates probabilistic predictions for Buy, Hold, and Sell actions. These predictions are aggregated by a central **teacher model** to produce the final investment recommendation.

4.4.1 Technical and Fundamental Analysis Model

The Technical and Fundamental Analysis Model analyzes the price behavior in the market and the financial stability of the firm. The technical factors are used to identify the trends in the market and the momentum of the firm. The technical factors are calculated on the basis of historical price changes and trading volumes in the market.

Apart from the technical factors, the model also analyzes the financial stability of the firm on the basis of various factors such as revenue growth, profitability, leverage, and others. The combined analysis of both the factors provides an overview of whether the firm is financially stable and whether there are good market conditions to invest in the firm.

The architecture and working of the combined approach to financial analysis are depicted in Figure 4.3.

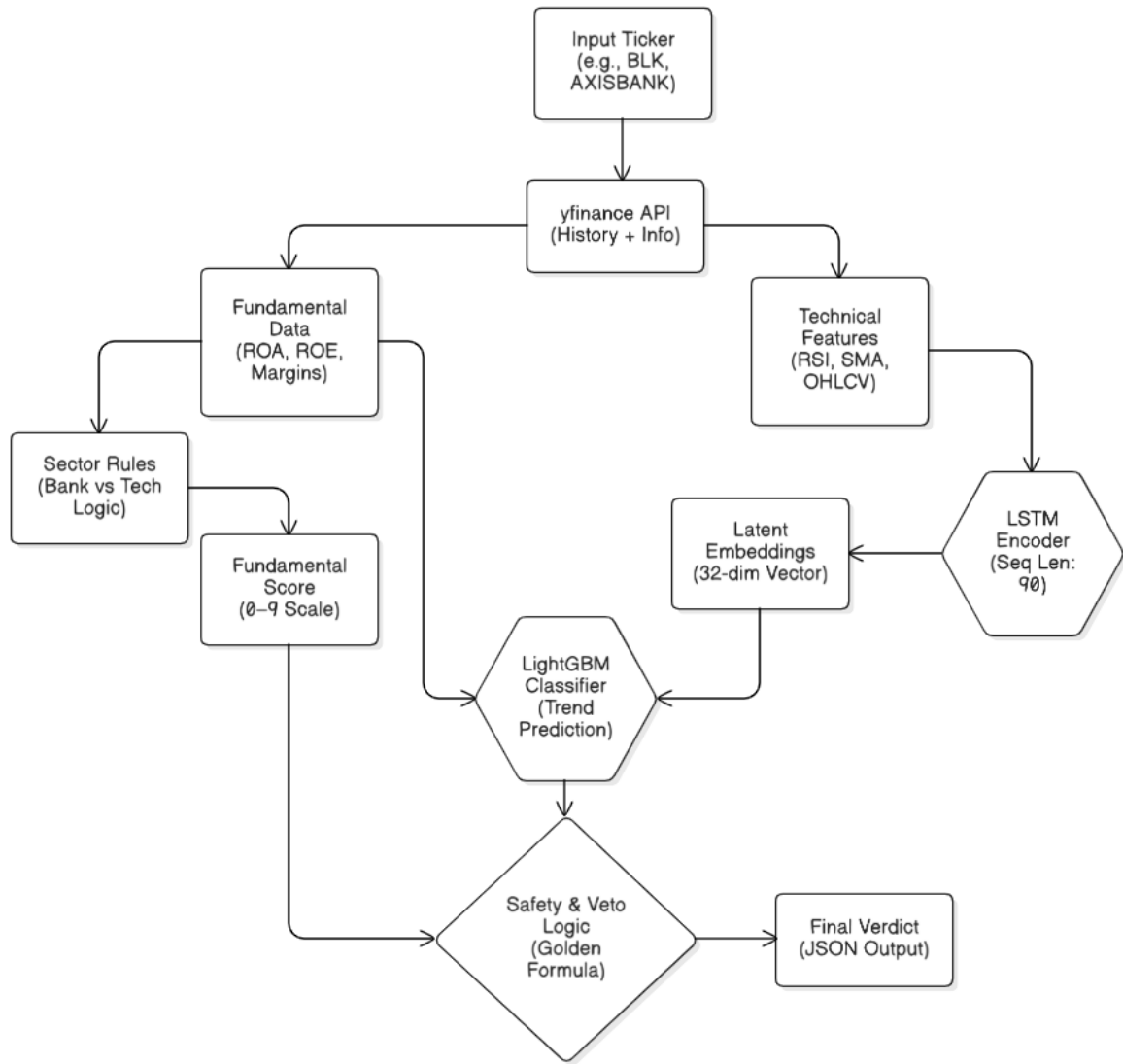


Figure 4.3: Architecture of the Technical and Fundamental Analysis Model

4.4.2 Sentiment Analysis Model

The Sentiment Analysis Model analyzes financial news articles, analyst opinions, and social media to determine the overall sentiment of investors towards a specific company. This is achieved by utilizing transformer models in natural language processing to classify textual data and determine whether the sentiment is positive, negative, or neutral in nature.

The various sources of information are utilized to determine the final sentiment score on the basis of factors such as relevance, credibility, and recency. This allows the system to determine the real-time market perception and investor behavior, which is an essential

factor in investment decisions. The architecture and processing flow of the sentiment evaluation mechanism are illustrated in Figure 4.4. This figure demonstrates the various textual data sources that are analyzed to determine the final sentiment.

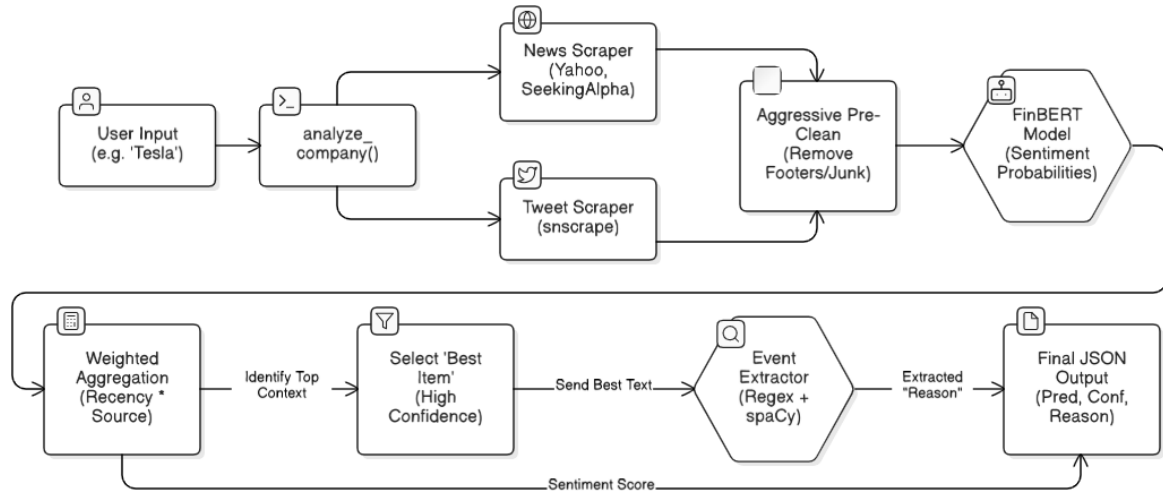


Figure 4.4: Architecture of the Sentiment Analysis Model

4.4.3 Alternative and Web Data Model

The Alternative and Web Data Model utilizes non-traditional market data to help interpret the signals received from various online platforms. The non-traditional data signals include social media signals, developer signals, hiring signals, search engine signals, and others related to the company.

The Alternative and Web Data Model analyzes the non-traditional data signals to identify the emerging public attention and market movements. The signals received from the non-traditional data sources are processed to derive a composite alternative data score, which represents the level of interest in the market related to the stock. The overall architecture of the data processing workflow of the Alternative and Web Data Model is depicted in Figure 4.5.

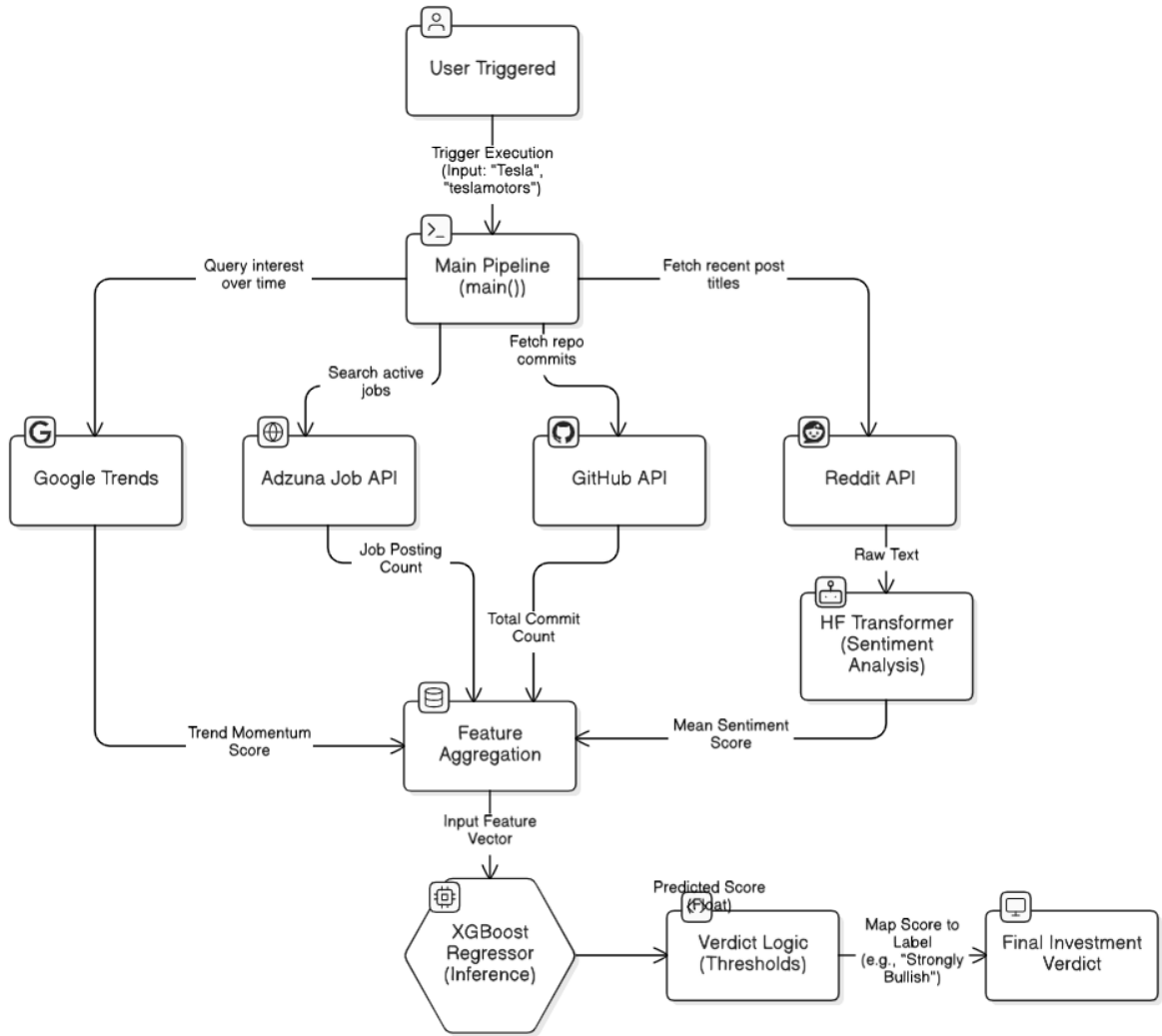


Figure 4.5: Architecture of the Alternative and Web Data Model

4.4.4 Macroeconomic Analysis Model

The Macroeconomic Analysis Model is used to assess various macroeconomic indicators that play a significant role in determining the overall performance of equity markets. It focuses on analyzing various macroeconomic indicators such as economic growth measured by GDP, inflation rates, and unemployment rates to determine if the economic environment is conducive or not to investments in equity markets. It is a very important tool that adds context to various company-level predictions, which is necessary to ensure accurate investment recommendations. The architecture of the macroeconomic evaluation is shown in Figure 4.6.

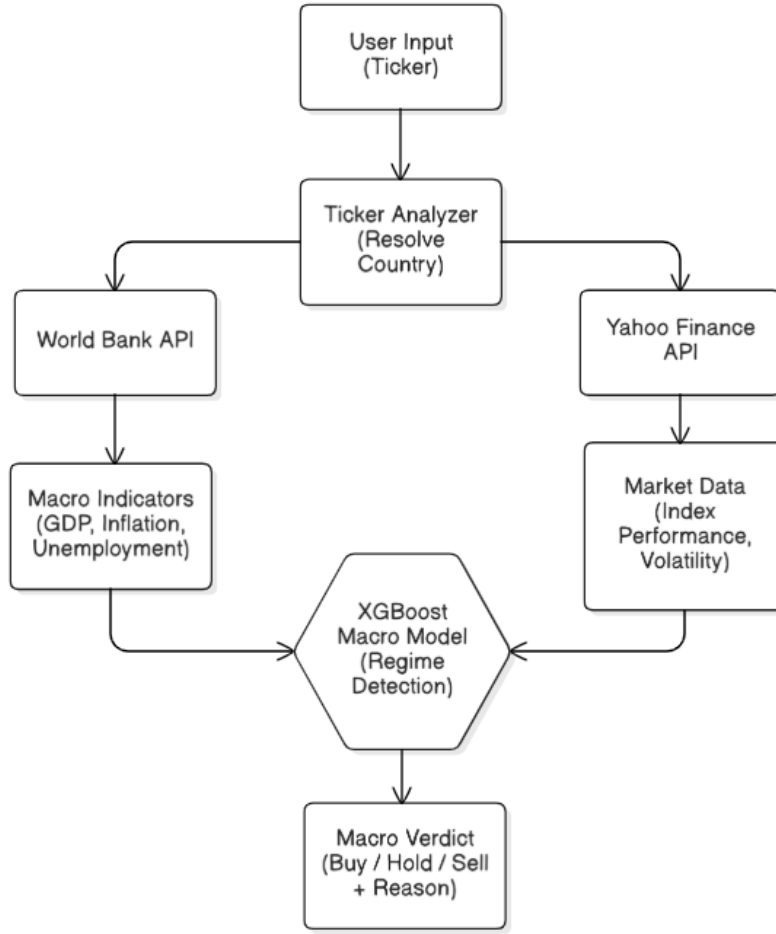


Figure 4.6: Architecture of the Macroeconomic Analysis Model

4.4.5 Market Cycle Model

The Market Cycle Model is a technique for analyzing the current financial market phase by carefully examining the current phase of the financial market based on key structural factors. The key structural factors include the strength of the long-term trends, the current implied financial market volatility indicated by the VIX index, and the current bond yield. The Market Cycle Model is able to identify whether the current financial market is experiencing an expansion phase, a contraction phase, or a neutral phase based on the current trends of the key structural factors.

Understanding the prevailing stage of the market cycle enables investors to align their investment strategies with broader market trends and risk conditions. The architectural design and decision-making flow of this model are illustrated in Figure 4.7, demonstrating how multiple market indicators are integrated to determine the current market phase.

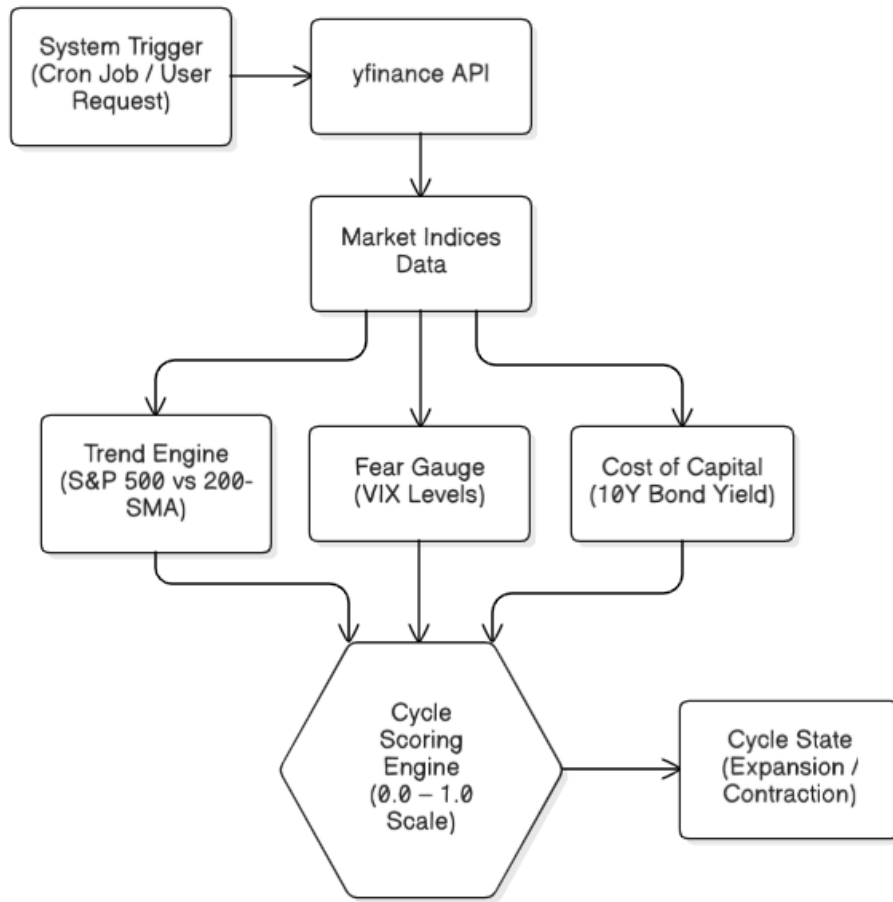


Figure 4.7: Architecture of the Market Cycle Model

4.4.6 Expert Consensus Model

The Expert Consensus Model integrates investment recommendations provided by professional financial analysts to capture market expertise and institutional perspectives. These recommendations are typically expressed as rating categories such as Strong Buy, Buy, Hold, Sell, and Strong Sell.

To enable quantitative analysis, the model converts these qualitative ratings into a weighted numerical score and computes a probabilistic distribution across Buy, Hold, and Sell outcomes. In addition, the total number of analysts contributing recommendations is considered to assess the reliability and confidence of the consensus signal. The structural design and aggregation process of the model are illustrated in Figure 4.8, highlighting how analyst opinions are transformed into actionable investment insights.

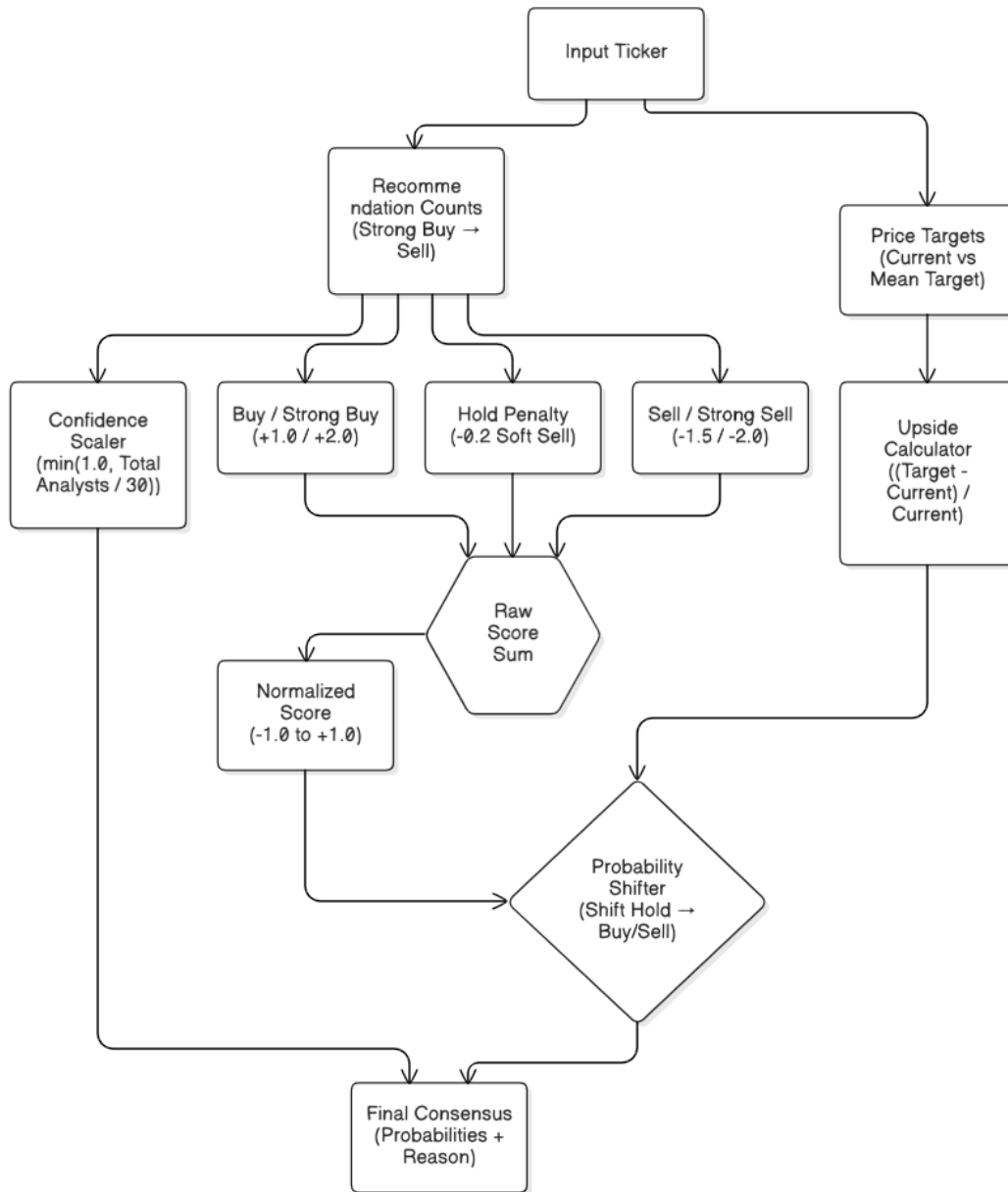


Figure 4.8: Architecture of the Expert Consensus Model

4.4.7 Teacher Model (Decision Aggregation)

The Teacher Model is the main decision-making part of the Veridian system. It uses a reliability-weighted aggregation method to combine the probability distributions produced by all the student models.

The student models produce a probability distribution for the decisions of Buy, Hold, and Sell, along with a confidence score. The teacher model uses reliability weights to calculate the final probability distribution.

The final investment recommendation is determined by selecting the action with the highest aggregated probability. The teacher model also generates an overall confidence score representing the stability and reliability of the recommendation.

4.5 Tools and Technologies

The development of Veridian utilizes modern web technologies and machine learning frameworks to ensure efficiency and scalability.

- **React.js:** Used for building the frontend interface and interactive dashboards.
- **Python:** Serves as the primary programming language for implementing machine learning models and backend services.
- **FastAPI:** Provides high-performance APIs for handling user queries and system communication.
- **Machine Learning Libraries:** Libraries such as Scikit-learn and PyTorch are used for implementing predictive models and data analysis algorithms.
- **Natural Language Processing Models:** Transformer-based models such as FinBERT are used for sentiment analysis of financial news.
- **Firebase / Cloud Storage:** Used for managing authentication, storing user data, and maintaining system logs.

4.6 Work Breakdown and Responsibilities

The development of the Veridian system was divided among team members based on functional modules to ensure efficient project execution.

- **Frontend Development:** Responsible for designing the user interface and implementing interactive dashboards.
- **Backend Development:** Responsible for developing APIs, managing server-side logic, and integrating machine learning models.

- **Machine Learning Model Development:** Responsible for implementing student models for financial analysis and designing the ensemble teacher model.
- **Data Processing and Integration:** Responsible for collecting financial datasets, performing preprocessing, and ensuring seamless data flow within the system.
- **System Testing and Documentation:** Responsible for validating system functionality and preparing technical documentation.

4.7 Hardware Requirements

The hardware requirements for developing and running the Veridian financial advisory system are listed below:

- Laptop or Desktop Computer
- Minimum 8 GB RAM (16 GB recommended)
- Intel i5 / AMD Ryzen 5 processor or higher
- Minimum 10 GB free storage
- Stable Internet connection
- Google Colab for cloud-based GPU support

4.8 Software Requirements

4.8.1 Programming Environment

- Python 3.x
- Google Colab
- Visual Studio Code (VS Code)

4.8.2 Machine Learning and Data Processing Libraries

- NumPy
- Pandas
- Scikit-learn
- XGBoost
- PyTorch

4.8.3 Natural Language Processing and Explanation Module

- HuggingFace Transformers
- Google FLAN-T5 (for explanation generation)

4.8.4 Backend and Database

- FastAPI
- Firebase Authentication
- Firebase Firestore

4.8.5 Data APIs

- Yahoo Finance API (Market and Fundamental Data)
- World Bank API (Macroeconomic Data)
- News APIs / RSS Feeds (Sentiment Data)

4.9 Project Timeline

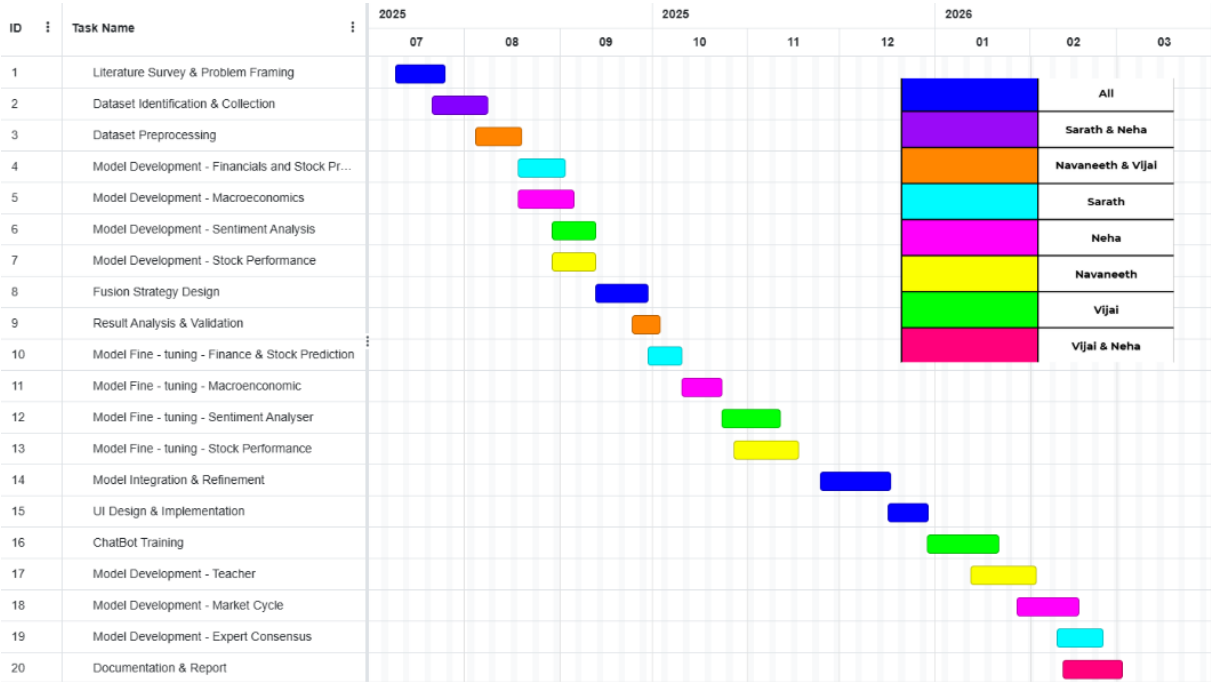


Figure 4.9: Project Timeline (Gantt Chart) of Veridian

4.10 Summary of Chapter

This chapter described the system design and architecture of the Veridian platform. The chapter presented the overall system architecture, component design, and module division of the system. The technologies used for development were also discussed, along with the work distribution among team members. Finally, the chapter outlined the key project deliverables and the development timeline followed during the implementation of the system.

Chapter 5

Results and Discussions

5.1 Overview

In this chapter, the results derived from the development and assessment of Veridian, a financial advisor system based on AI developed by Veridian Corp., will be presented in detail. Evaluation objectives are aimed at examining the degree of success that the teacher-student ensemble architecture used within Veridian may achieve when it comes to the generation of stable and easily interpretable investments in stocks made by both the individual user and the teacher-student ensemble architecture.

To complete this task, the Veridian system is designed to collect and process financial data from an assortment of sources, such as historical stock prices, financial indicators, macroeconomic indicators, news articles analysing companies, and alternative web-based signals. The above-mentioned inputs are inputted into each of the student models that exist within the Veridian system, and then the outputs from each of the student models are aggregated together by the teacher model to provide the final investment recommendation to either buy, hold or sell an investment.

The purpose of this chapter is to demonstrate the process that Veridian uses to analyse investment queries related to stock investments by processing an input query pertaining to stocks, using multiple financial models to analyse the query, and generating an easily interpretable set of investment insights. The output of the evaluations in this chapter includes model outputs with input and output for each student model, system predictions, and snapshots of the user interface in Veridian to demonstrate the capabilities of the current system that has been implemented. As part of this chapter, the performance of the current version of Veridian will also be discussed in relation to decision-making stability, ease of understanding, and the integration of heterogeneous financial signals in an overall investment recommendation framework.

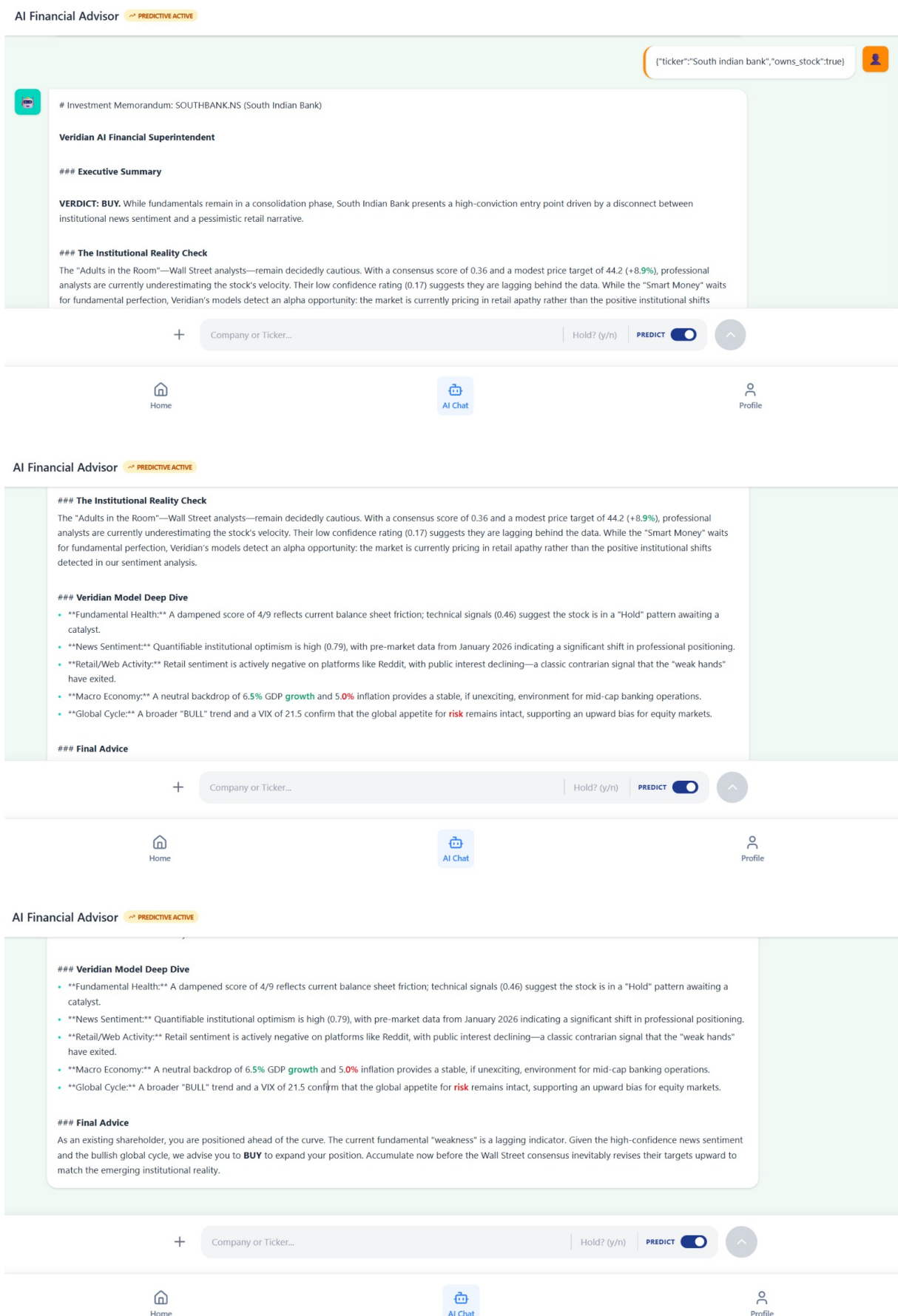
5.2 Results

Figures below depicts examples of the types of investment recommendation results generated from Veridian’s system for different queries against stocks that were submitted to the Chatbot interface.

When a user submits a request by inputting either a company’s name or its ticker symbol, the system performs an analysis using its teacher-student ensemble architecture, with multiple analytical models assessing each of several financial indicators, including: market sentiment, technical trend data, macroeconomic factors, alternative data, and expert opinion.

As a result, the teacher model collects the probabilistic outputs (the number-crunching) of the different models to create a final investment recommendation and the final recommendation is categorized by type (Buy, Hold, or Sell); Each recommendation is supported by several reasoning or insight examples that enable the user to have an understanding of how the analytical signals led to the final recommendation.

The screenshots presented below demonstrate how the Veridian chatbot worked with users as they made queries and provided the user with structured, analyzable investment insights regarding the request; The various investment outputs highlight Veridian’s ability to generate understandable investment recommendations; In effect, the recommendations from the Veridian system encourage the investor to use multiple analytical indicators when making an investment decision as opposed to only using a single indicator.



5.3 Output

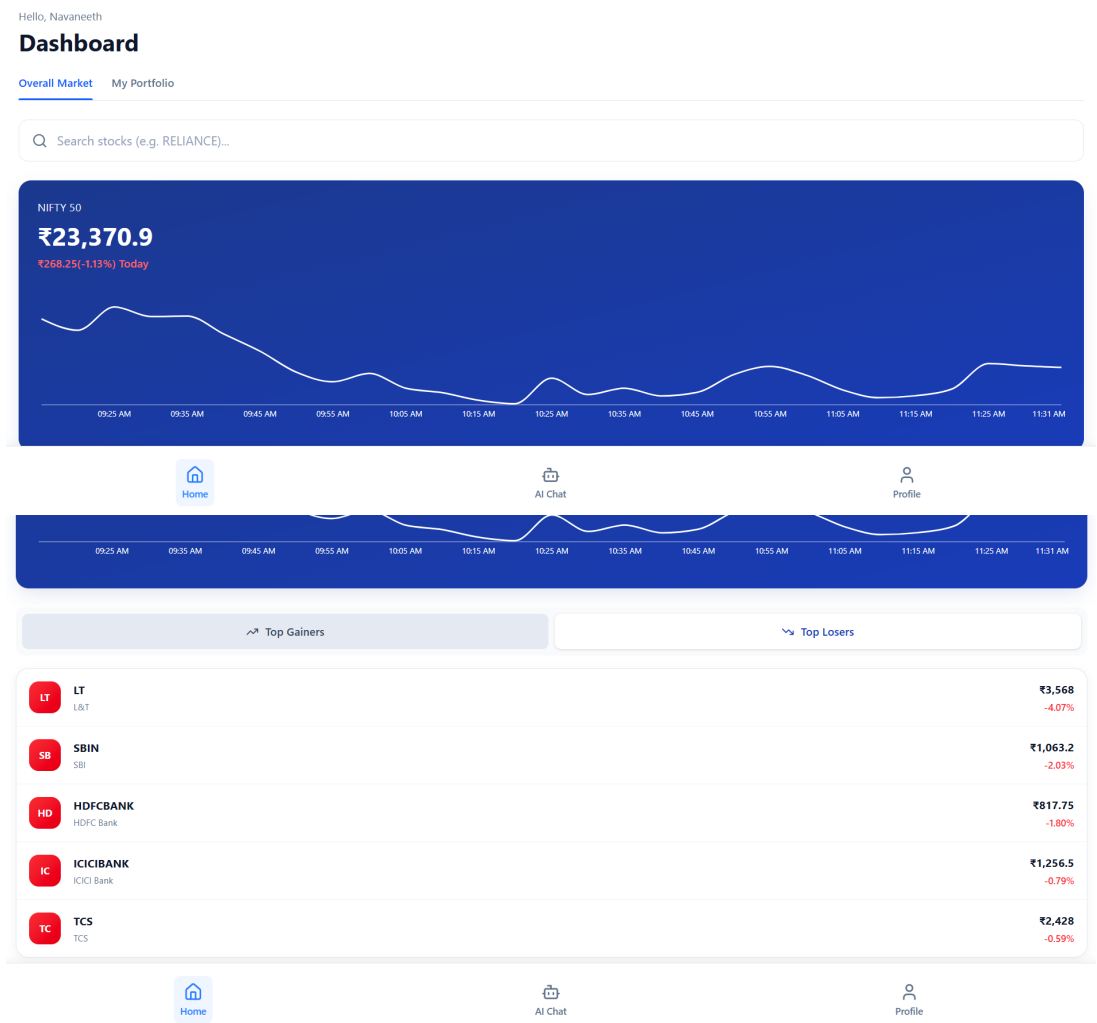


Figure 5.2: Veridian Dashboard Interface

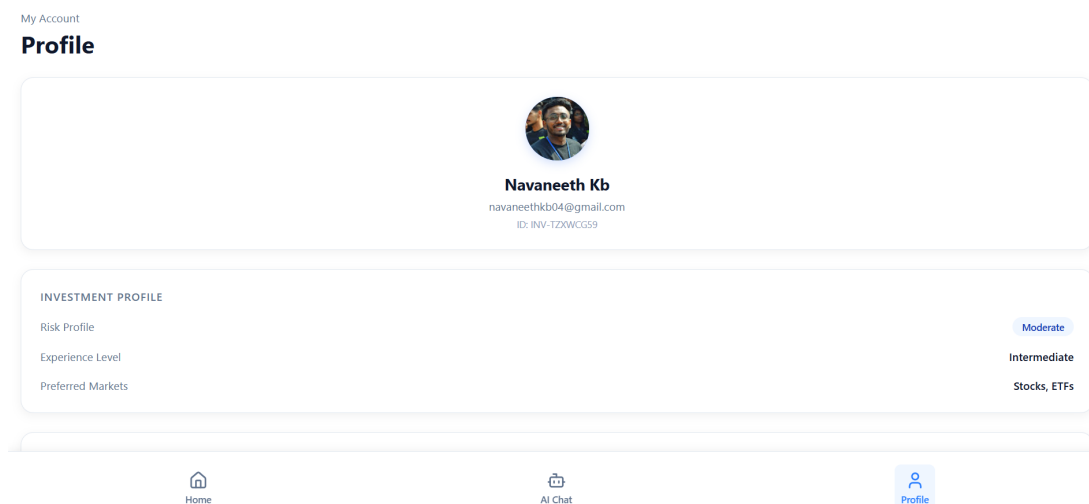


Figure 5.3: User Profile Management Interface

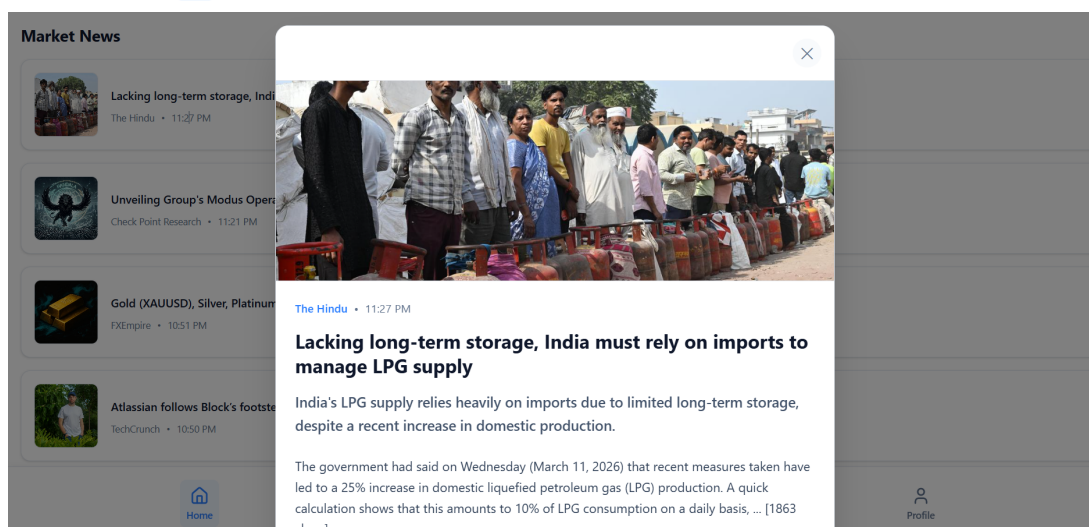
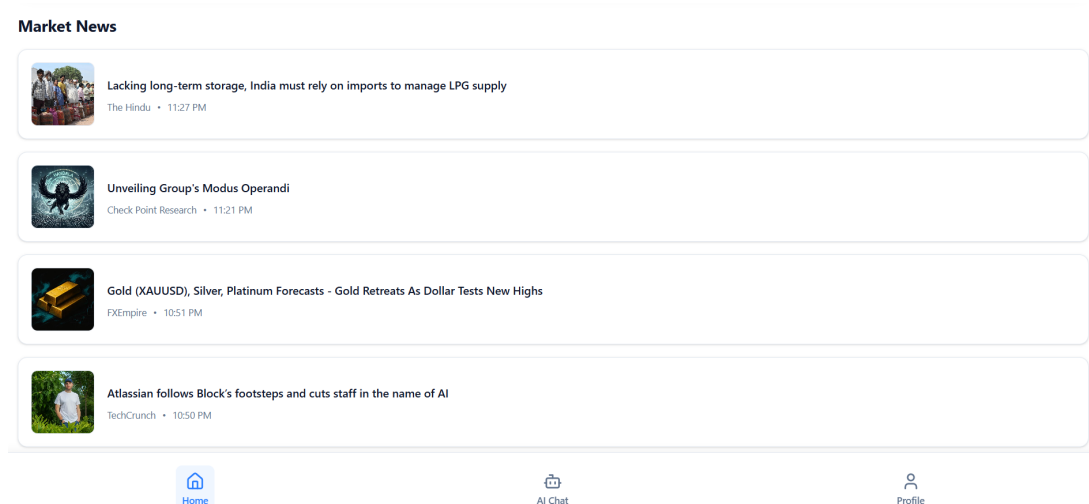


Figure 5.4: Latest Financial News

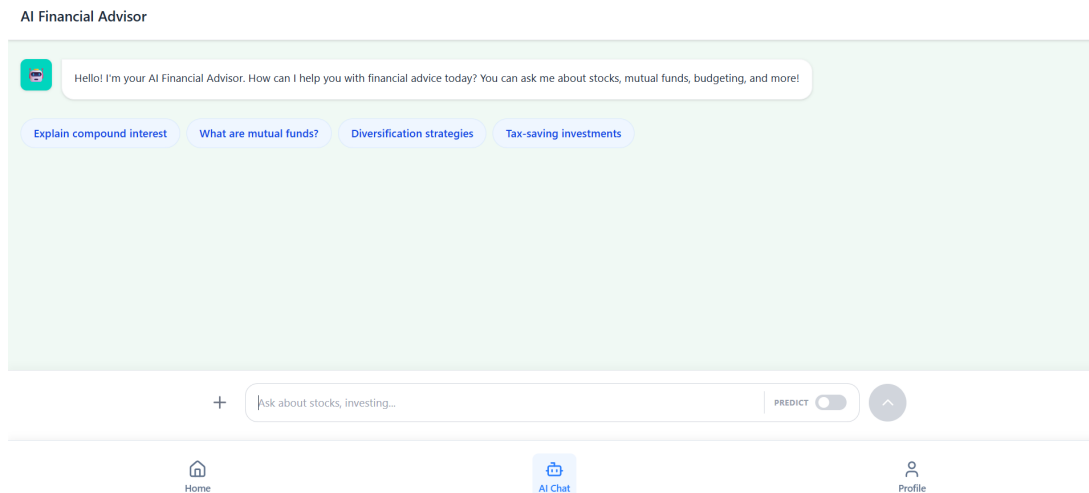


Figure 5.5: Chatbot Interface for Stock Query Interaction

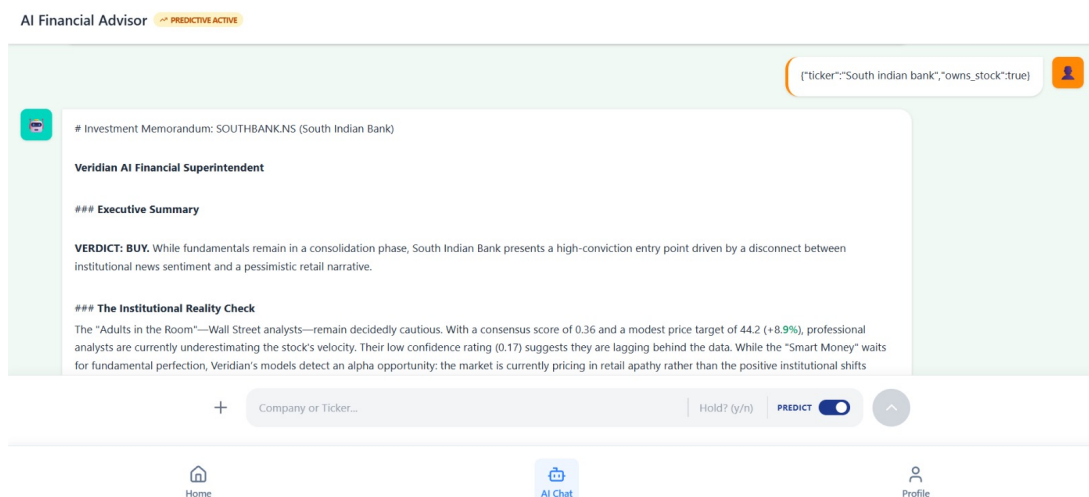


Figure 5.6: Investment Recommendation Output Screen

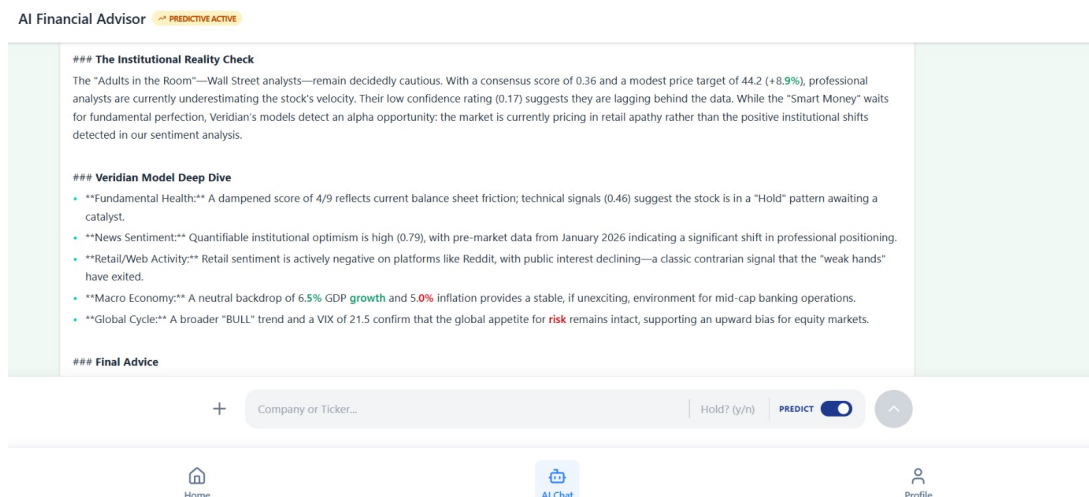


Figure 5.7: Explanation and Model Contribution Visualization

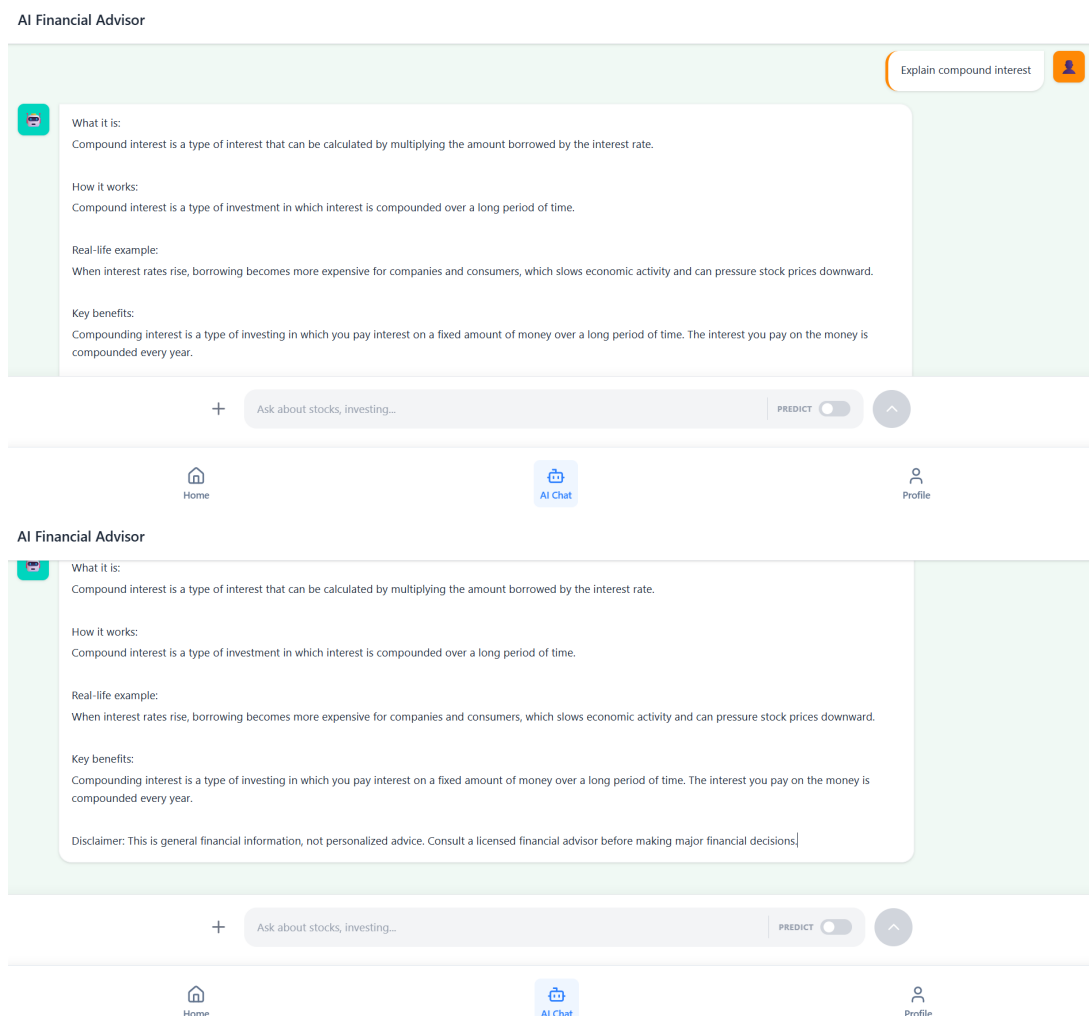


Figure 5.8: Financial Query Output Screen

5.4 Discussion

The results derived from the Veridian system show the effectiveness of the teacher–student architecture for evaluating financial decisions. By combining predictions from multiple analytical models, the system can evaluate a wide range of financial metrics and develop consistent methods for investing.

One major benefit of the proposed system is its ability to interpret and reconcile multiple signals from different analytical models with conflicting outcomes. For example, sentiment-based models can tend to react to events in the news on a short-term basis, whereas long-term fundamentals may be more focused on a security’s long-term performance. The teacher model uses weighted averages of reliability to combine the potential signals from models with incompatible performance into a single, interpretable recommendation.

Another key factor in this system is utilizing a confidence score associated with each possible prediction. This confidence score allows the end-user to determine how confident they should be in the model based upon the level of agreement between multiple analytical models. A high confidence score indicates significant agreement between the respective models, while a low confidence score indicates there was a lack of consensus among the models being evaluated, and the signal(s) involved are therefore uncertain or contradictory.

Veridian’s user interface shows how easy the platform is to use. Beyond just being able to ask a question about a specific stock via a chatbot, users can also do a variety of things with their query results, including getting structured investment information as well as reports that explain their results. In addition, the various visual tools added to the platform improve overall transparency and provide additional help for users when trying to understand why they received specific recommendations.

The overall results of the study indicate that the proposed system successfully integrates many different types of financial datasets and produces consistent investment recommendations appropriate for decision support.

5.5 Summary

This chapter contained the outcomes from the deployment of the Veridian AI-based financial consulting software. The results demonstrate how well the teacher-student ensemble system can evaluate many financial metrics at once and create intelligible investment suggestions.

In addition to the use of several different analyses to increase decision-making reliability and minimize the potential risks associated with using only one financial metric, they can create greater confidence in and improve clarity of the recommendations and predictions by using both a confidence metric and an explanation process.

The output images demonstrate several of the functions available in the developed application, such as chatbot interaction, prediction dashboards, and report generation. Collectively, the evidence supports the conclusion that the proposed system can effectively operate as an AI-supported financial decision support system.

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

The Veridian system presents an opportunity to integrate different technologies for use in financial advisory systems by utilizing various types of analytical approaches (e.g., both technical and fundamental data), sentiment analysis, macroeconomic data sources, alternative data sources and experts' opinions on the data being analyzed. The process used to provide a more complete picture of the evaluation of stocks is called an “analytical framework” because it pulls together all available analytical sources to evaluate the stock and ultimately provide an investment recommendation (buy, hold, or sell).

The teacher-student architecture allows the models to analyze the financial data independently while the teacher aggregates the many models' predictions using a weighted aggregation strategy producing a recommendation. This allows for better stability in the recommendation by applying a “reliability-weighted” algorithm-based decision model to a pool of independent analyses.

The experimental results show that using an ensemble decision model by integrating the various heterogeneous financial outcomes from the various sources produced a decision model that has less variance in the results (greater stability) than the original model did when using a single-model analysis; additionally, it can improve the accuracy of recommendations through transparency (confidence scores) and a clear explanation of the data and methodology used to arrive at that specific recommendation. Ultimately, the Veridian system's ability to support AI-based decision support systems in assisting investors with making intelligent investment decisions from analyzing structured financial data indicates the underlying promise of AI as a decision support system.

6.2 Future Scope

Although Veridian presently serves as a strong framework for stock analysis, several avenues exist for augmenting its efficacy. A potential enhancement involves the incorporation of reinforcement learning techniques, which would facilitate the dynamic adjustment of model weightings predicated on real-world market performance. A further extension would allow for users to have customized investment strategies that take into account their individual risk profile and portfolio preferences. Incorporation of real-time data streaming and simulated trades will also enable analysis of how effective the system will be under fluctuating market conditions. Another avenue for improvement would come from expanding on what types of alternative data can be used in predicting stock trends — such as adding ESG measures or other global indicators into consideration when developing predictive models. Finally, the platform could be utilized as a cost-efficient scalable, cloud-based solution for delivering financial advice to greater numbers of potential customers.

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Appendix A: Presentation

VERIDIAN[®]

SEE BEYOND NUMBERS

GUIDE: ANU MARIA JOYKUTTY

VIJAI KIREN S - U2203209

NAVANEETH K BIJUMON - U2203150

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SARATH GIRISH - U2203190

DATE : 03/03/2026

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INTRODUCTION

- over 80 different factors to research before making an informed investment decision.
- Business model, revenue streams, company history, track record, products and services, market position, competitive advantage, management quality, leadership experience, corporate governance, mission and vision, P/E ratio, ROE, ROA, gross profit margin, net profit margin, operating margin, P/B ratio, P/S ratio, PEG ratio, enterprise value to EBITDA, debt-to-equity ratio, current ratio, quick ratio, cash flow analysis, working capital, historical financial performance, revenue growth trends, EPS growth, cash flow stability, balance sheet strength, tax liabilities, dividend history, industry growth potential, market size, competitive landscape, industry strengths, cyclical nature, barriers to entry, market share, beta coefficient, stock volatility, business risks, regulatory risks, currency risks, litigation history, environmental factors, expansion plans, new product pipeline, R&D investments, strategic partnerships, intellectual property, technology adoption, geographic expansion, stock price trends, trading volume, support and resistance levels, moving averages, chart patterns, corporate structure, ownership, material contracts, permits and licenses, insurance coverage, regulatory compliance, pending litigation, supply chain efficiency, manufacturing processes, quality control, customer satisfaction, employee retention, IT infrastructure, consensus revenue estimates, profit forecasts, analyst recommendations, price targets, institutional interest, environmental impact, social responsibility, board composition, executive compensation, transparency practices, stock options, insider trading, recent news, peer comparison, margin of safety calculation.

PROBLEM DEFINITION

Retail investors struggle to process large volumes of complex, fast-changing market data and often lack the expertise to interpret trends, assess risks, and identify opportunities. As a result, they frequently make decisions based on incomplete or outdated information, leading to poor portfolio performance, missed opportunities, and reduced confidence.

PROJECT OBJECTIVE

- Develop a data-driven financial advisory web-app.
- Provide AI/analytics-based investment recommendations
- Enable structured portfolio and P/L tracking.
- Improve financial literacy through intelligent insights
- Deliver a scalable and user-centric financial planning solution
- To integrate the latest financial news to support informed decision-making

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PURPOSE

- To simplify stock market investment for individual investors
- To provide a centralized platform for portfolio tracking and stock analysis
- To assist users in making informed investment decisions using data and news insights
- To bridge the gap between complex market data and everyday investors

NEED

- Many investors lack structured guidance in stock market decision-making
- Financial data and market news are scattered across multiple platforms
- Beginners face difficulty understanding market trends and stock performance
- There is a need for an integrated system that combines portfolio tracking, recommendations, and financial updates

PURPOSE AND NEED

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LITERATURE SURVEY

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Paper & Year	Focus & Contribution	Pros (what we can reuse)	Cons / Gaps (what we'll change)
Theate & Ernst 2021 – “Deep RL for Algorithmic Trading” (Expert Syst. Appl.)	Implements Deep Q-Network on US equities with reward shaping & transaction-cost penalties	- Practical deep-RL template with cost-aware reward - Publishes open-source code	- Overfits to 2010–2019 bull market; no crisis years - Lacks explainability → we'll add SHAP/attention heat-maps
Kwak et al. 2023 – “Self-Attention Deep DRL with Hybrid Loss” (Inf. Sci.)	Combines Transformers and recurrent nets for intraday futures signals	- Attention layer improves feature attribution - Hybrid loss balances profit & risk	- Uses minute data from one Korean index future only → we'll generalise across assets; include cross-market regularisation
Passalis et al. 2021 – News-enhanced finBERT + LSTM/CNN	Integrates sentiment embeddings with technical data for stock trades	- Shows measurable alpha from NLP sentiment - Modular: easy to plug new news feeds	- Only daily frequency; ignores execution cost → we'll extend to higher frequency & slippage modelling
Felizardo et al. 2022 – “Supervised vs RL in Crypto” (Expert Syst. Appl.)	Benchmarks RL bots against classical classifiers on BTC/ETH	- Provides strong baselines & rigorous cross-validation	- Crypto-only; no portfolio context → we'll add multi-asset allocation layer and risk budgeting

VERIDIAN²

8

PROPOSED METHOD

VERIDIAN  9

CORE IDEA

- Veridian uses a Teacher–Student Ensemble Architecture
- Multiple specialized AI models (students) analyze a stock independently
- A central Teacher Model aggregates their outputs using reliability-weighted averaging
- Final output: Buy / Hold / Sell recommendation with confidence score

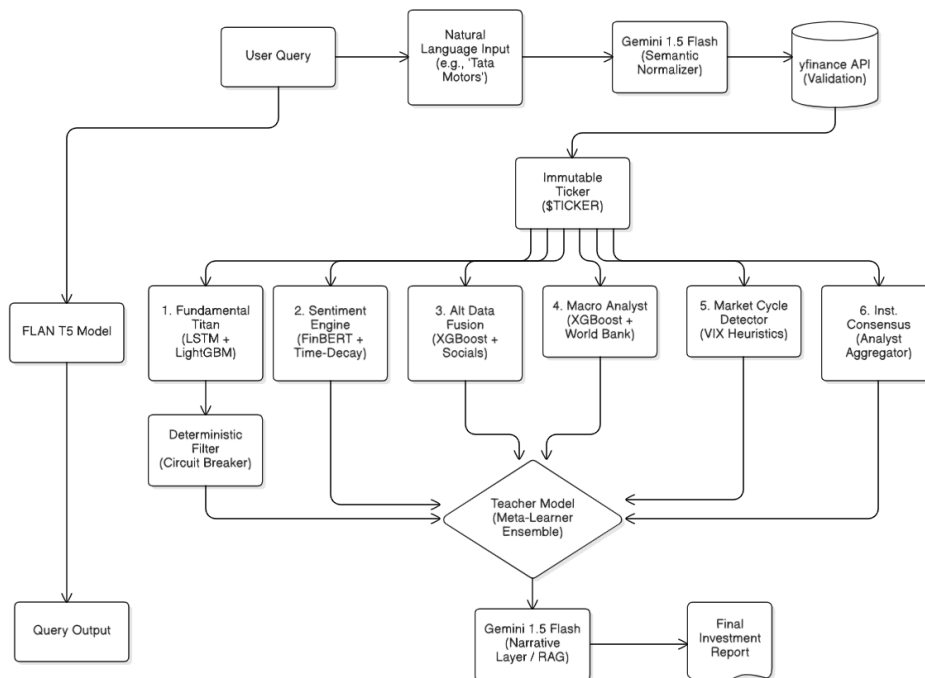
SYSTEM WORKFLOW

- User submits query via chatbot
- If Prediction Toggle OFF → General AI model responds
- If Prediction Toggle ON → Stock prediction pipeline activates
- Company name resolved to standardized ticker symbol
- Automated multi-source data collection & preprocessing
- Student models run in parallel (Buy / Hold / Sell probabilities)
- Teacher model performs reliability-weighted aggregation
- Final recommendation + confidence + explanation delivered

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ARCHITECTURE DIAGRAM

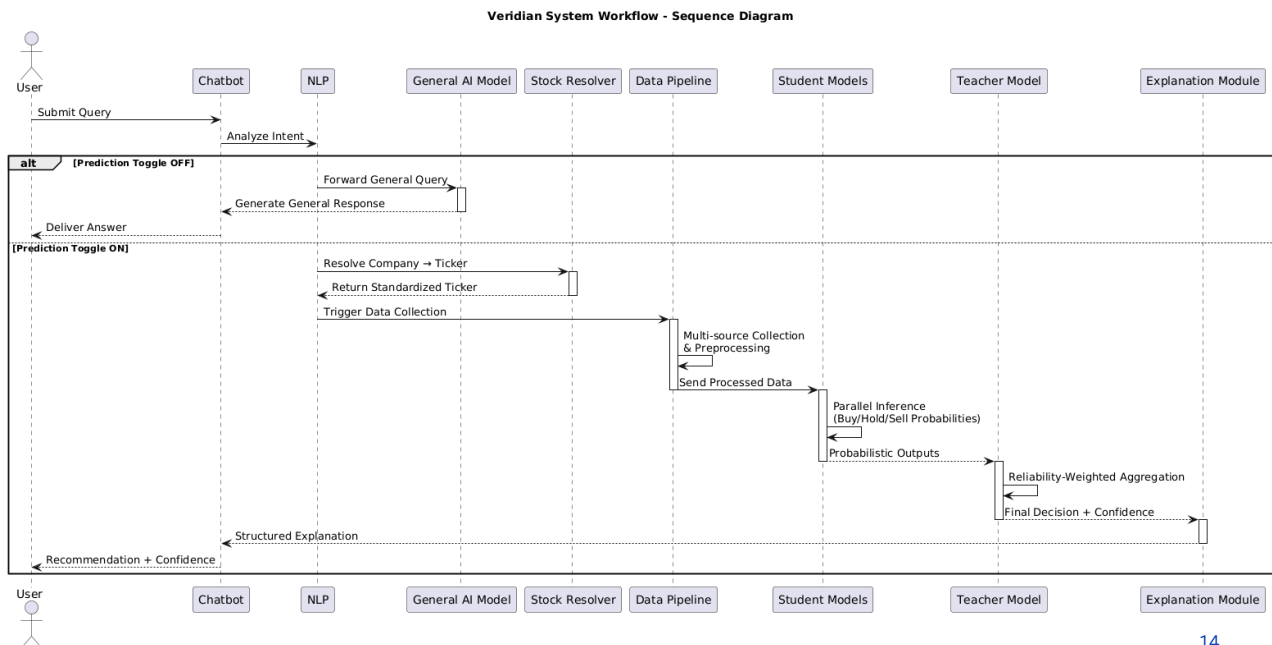
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SEQUENCE DIAGRAM

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VERIDIAN 14

MODELS

Fundamental Titan
(LSTM + LightGBM)
Equity Research Analyst

Sentiment Engine
(Finbert)
Market Sentiment Analyst

Alternative Data Fusion
(XGBoost)
Alternative Data Strategist

Macro Analyst
(XGBoost)
Macroeconomic Analyst

Market Cycle Detector
(Heuristic Model)
Risk & Volatility Strategist

Institutional Consensus
(Analyst Aggregator)
Institutional Research Desk

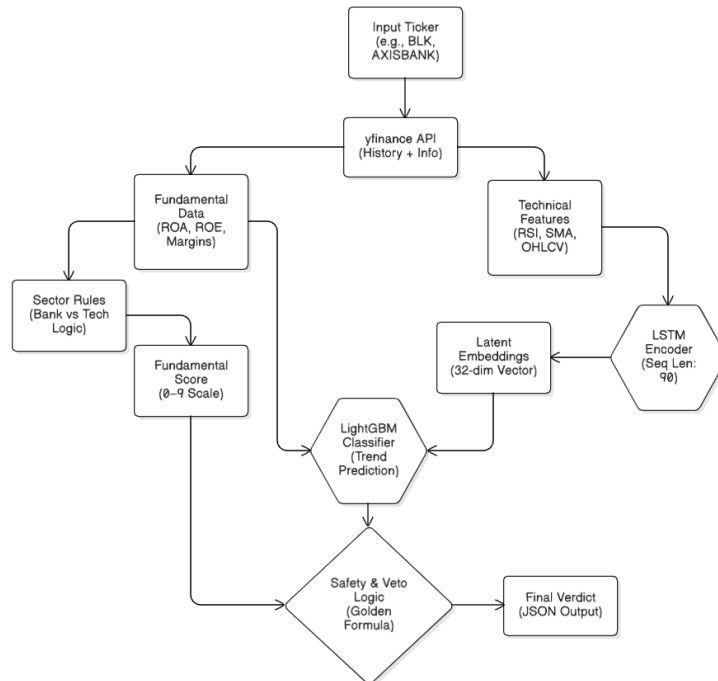
Teacher Model
(Random Forest Meta - Learner)

VERIDIAN 15

TECHNICAL + FUNDAMENTAL MODEL

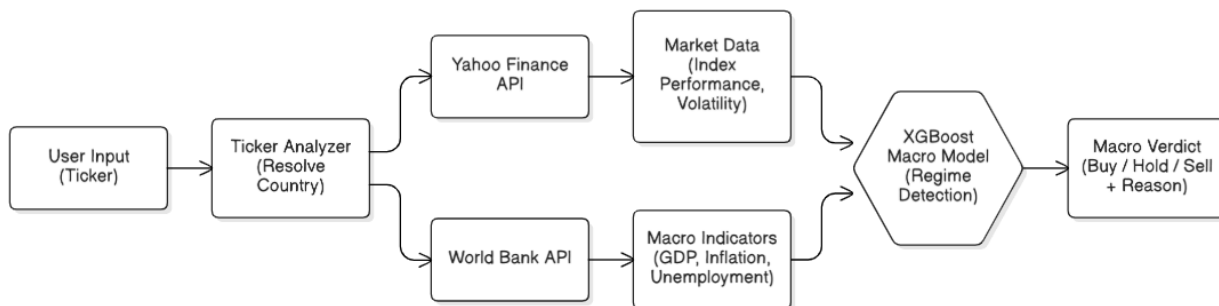
- Ingestion: Fetches data via yfinance.
- Feature Engineering: Calculates RSI, 50-day Moving Average, and "Synthetic P/E."
- LSTM Processing: Turns 90 days of price history into a pattern vector.
- Fundamental Scoring: Scores the company's health based on its specific sector.
- Weighted Fusion: Combines Technical Health (60%) and Fundamental Health (40%).
- Veto Check: Applies "Bearish Bias" rules to prevent buying into a falling knife.

VERIDIAN 16



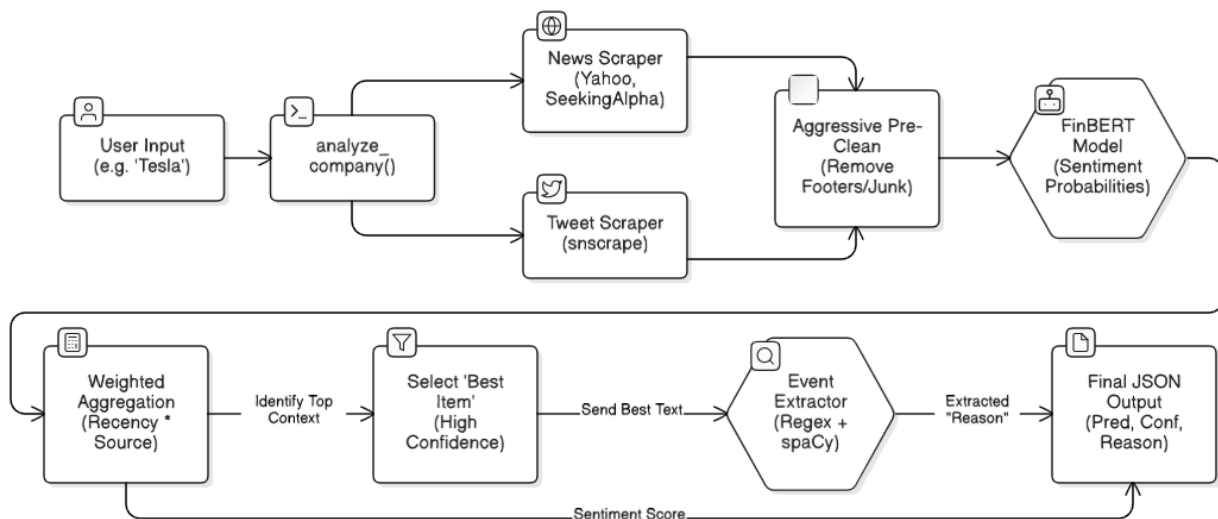
MACROECONOMIC MODEL

- Uses GDP growth, inflation, and unemployment to capture overall economic conditions
- Macroeconomic data sourced from the World Bank and mapped to each stock's country
- Yearly economic data converted to monthly format
- XGBoost classifier trained on macro indicators with time-series split
- Predicts BUY / SELL / HOLD based on 3-month forward returns
- Outputs confidence score and clear economic reasoning for explainability



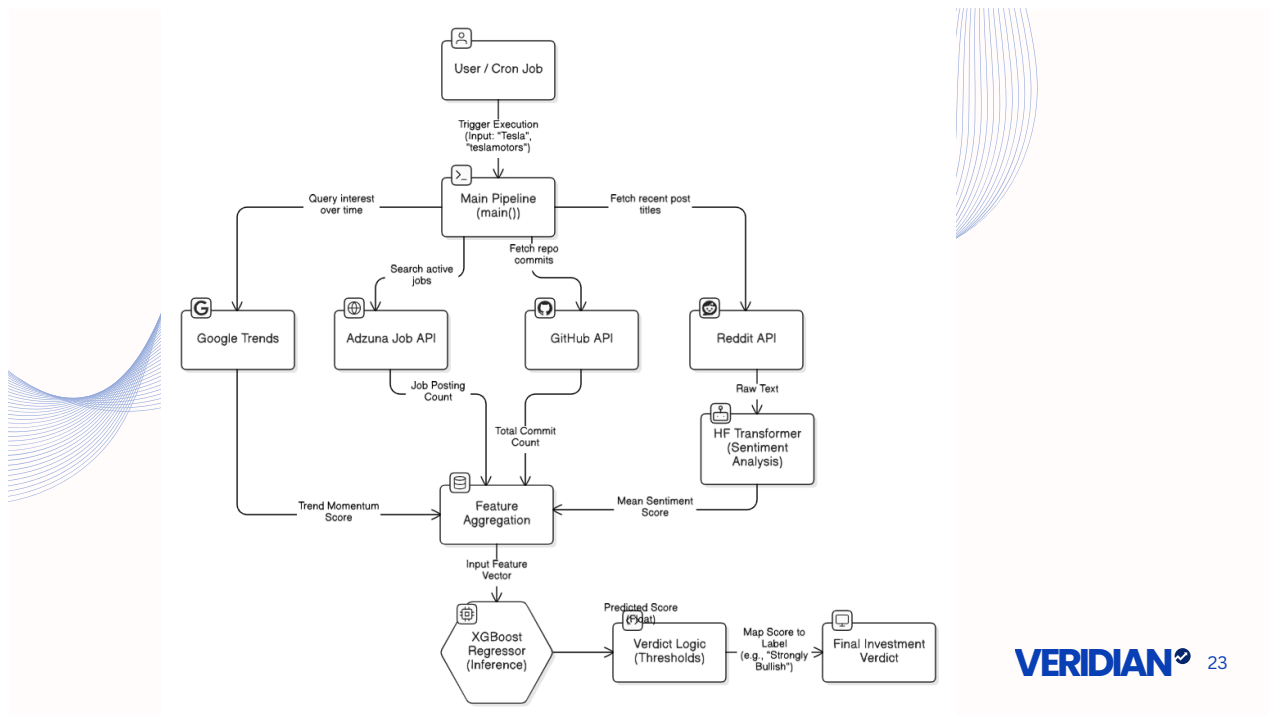
NEWS & SENTIMENT MODEL

- This Pipeline fuses Transformer-Based Sentiment Analysis (FinBERT) for quantifying market emotion with Automated Event Extraction (spaCy) to identify the root cause of volatility.
- It synthesizes raw probability scores into a Context-Aware Investment Verdict, validating numerical signals with concrete "Real-World Triggers" such as earnings beats or leadership changes.



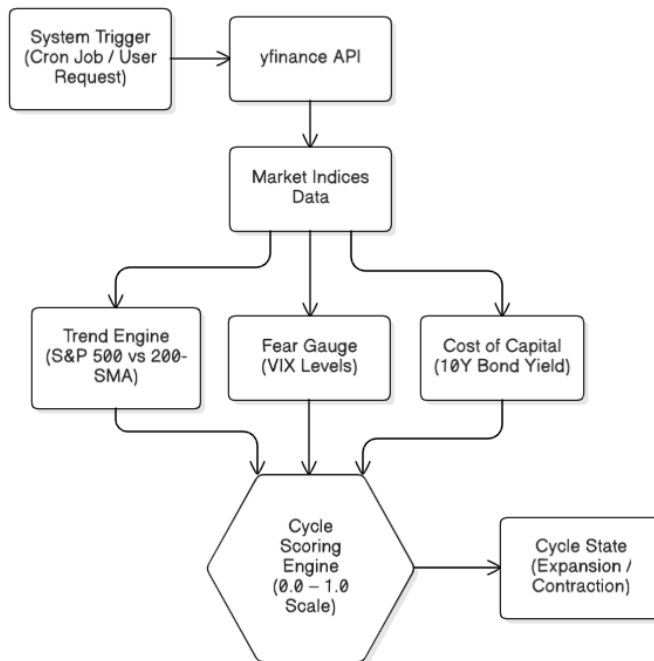
ALTERNATE & WEB DATA MODEL

- This Model fuses Social Sentiment & Search Momentum (Reddit + Google Trends) for tracking viral market interest with Operational Growth Metrics (Hiring + GitHub Activity) to validate business health.
- It synthesizes these diverse signals via XGBoost into a unified Investment Confidence Score, recommending entries only when "Public Hype" is backed by verifiable "Corporate Expansion."



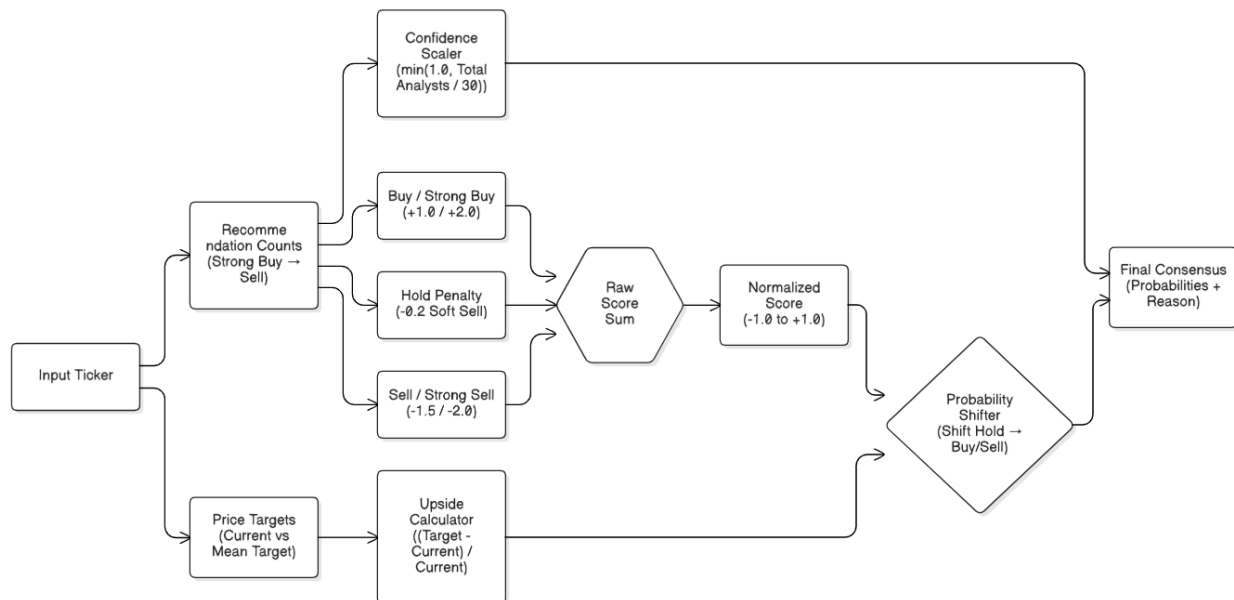
MARKET CYCLE ANALYST

- Role: The "Market Weather Station." It determines if the overall economic environment is safe for investing.
- Logic: Uses a heuristic scoring engine (0.0 - 1.0) to detect Risk-On vs. Risk-Off regimes.
- Key Signals:
- Trend: Checks if the S&P 500 is above/below its 200-day average (Bull vs. Bear).
- Fear: Monitors the VIX to detect panic (>30) or complacency (<12).
- Cost of Capital: Tracks 10-Year Treasury Yields to gauge borrowing pressure.
- Output: A Multiplier Score that dampens or vetoes "Buy" signals during market crashes.



ANALYST CONSENSUS ENGINE

- Role: The "Voice of Wall Street." Aggregates professional institutional ratings to gauge smart money sentiment.
- Logic: Uses a Weighted Scoring System that corrects for inherent analyst optimism bias:
- Strong Buy: +2.0 (High Conviction)
- Hold: -0.2 (Treated as a "Soft Sell" / Underperform)
- Sell: -1.5 (Penalized heavily as analysts rarely issue Sells)
- Key Features:
- Upside Check: Compares the Current Price against the Average Analyst Target to calculate potential profit %.
- Volume Confidence: Trust score scales with coverage depth (e.g., 30+ analysts = 100% Confidence; 3 analysts = 10% Confidence).



TEACHER MODEL

Inputs from Student Models

- Buy / Sell / Hold prediction
- Confidence score
- Model-specific rationale

Meta-Feature Engineering

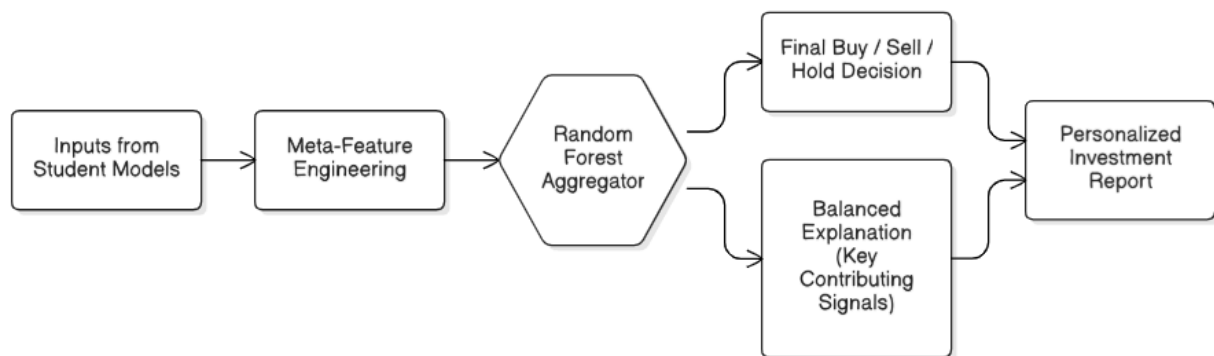
- Encoded trading signals
- Confidence values
- Sentiment & trend indicators
- Historical performance metrics
- Converts heterogeneous outputs into a structured feature vector.

Random Forest Aggregation

- Learns non-linear relationships
- Adapts to different market conditions
- Handles conflicting opinions
- Reduces overfitting via bagging
- Learns when to trust which model (not simple averaging)

Output Generation

- Final Buy / Sell / Hold decision
- Balanced explanation
- Key contributing signals highlighted
- Personalized report via Gemini



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RESULTS & OUTPUTS

VERIDIAN[®] 31

A decorative graphic consisting of multiple thin, overlapping blue lines that form a series of smooth, undulating waves across the middle half of the page.

STOCK 1: FINANCE

VERIDIAN[®] 32

Veridian output:

VERIDIAN

AI Financial Advisor PREDICTIVE ACTIVE

("ticker":"South indian bank","owns_stock":true)



Investment Memorandum: SOUTHBANK.NS (South Indian Bank)

Veridian AI Financial Superintendent

Executive Summary

VERDICT: BUY. While fundamentals remain in a consolidation phase, South Indian Bank presents a high-conviction entry point driven by a disconnect between institutional news sentiment and a pessimistic retail narrative.

The Institutional Reality Check

The "Adults in the Room"—Wall Street analysts—remain decidedly cautious. With a consensus score of 0.36 and a modest price target of 44.2 (+8.9%), professional analysts are currently underestimating the stock's velocity. Their low confidence rating (0.17) suggests they are lagging behind the data. While the "Smart Money" waits for fundamental perfection, Veridian's models detect an alpha opportunity: the market is currently pricing in retail apathy rather than the positive institutional shifts



Company or Ticker...

Hold? (y/n)

PREDICT ON



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AI Chat



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AI Financial Advisor PREDICTIVE ACTIVE

The Institutional Reality Check

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Veridian Model Deep Dive

- **Fundamental Health:** A dampened score of 4/9 reflects current balance sheet friction; technical signals (0.46) suggest the stock is in a "Hold" pattern awaiting a catalyst.
- **News Sentiment:** Quantifiable institutional optimism is high (0.79), with pre-market data from January 2026 indicating a significant shift in professional positioning.
- **Retail/Web Activity:** Retail sentiment is actively negative on platforms like Reddit, with public interest declining—a classic contrarian signal that the "weak hands" have exited.
- **Macro Economy:** A neutral backdrop of 6.5% GDP growth and 5.0% inflation provides a stable, if unexciting, environment for mid-cap banking operations.
- **Global Cycle:** A broader "BULL" trend and a VIX of 21.5 confirm that the global appetite for risk remains intact, supporting an upward bias for equity markets.

Final Advice



Company or Ticker...

Hold? (y/n)

PREDICT ON



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AI Chat



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Veridian Model Deep Dive

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- **Global Cycle:** A broader "BULL" trend and a VIX of 21.5 confirm that the global appetite for risk remains intact, supporting an upward bias for equity markets.

Final Advice

As an existing shareholder, you are positioned ahead of the curve. The current fundamental "weakness" is a lagging indicator. Given the high-confidence news sentiment and the bullish global cycle, we advise you to **BUY** to expand your position. Accumulate now before the Wall Street consensus inevitably revises their targets upward to match the emerging institutional reality.



Company or Ticker...

Hold? (y/n)

PREDICT ☒



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AI Chat

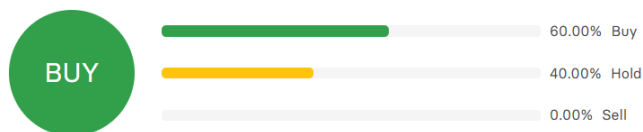


Profile

Analysts rating for South Indian Bank Ltd derived from IND Money:

Analyst Rating for South Indian Bank Ltd stock

based on 5 analysts



Based on 5 analysts offering long term price targets for South Indian Bank Ltd. An average target of ₹47.5

Source: Reuters LSEG Refinitiv

VERIDIAN 37



— — —

The "Adult in the Room"—a consensus of 33 Wall Street analysts—maintains an exceptionally high confidence level (1.00) in the stock's upward trajectory, setting an

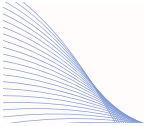


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PREDICT ☒

Profile

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AI Financial Advisor PREDICTIVE ACTIVE

- reporting.
- **Technical Signal**: Score 0.84 (Buy). Strong price action and trend indicators suggest the path of least resistance remains upward.
 - **News Sentiment**: High-conviction positive sentiment (0.82) centered on the narrative that Bharti's solid earnings quality is not yet fully appreciated by the retail sector.
 - **Retail/Web Activity**: Interest is rising (0.23 confidence). We are in the early stages of public "hype," providing a window to increase exposure before retail momentum spikes.
 - **Macro & Global Cycle**: Neutral domestic macro (6.5% GDP) paired with a Global Bull trend. A VIX of 21.4 indicates manageable volatility within a broader expansionary phase.

Final Advice

Current Status: Already Owned.

As an existing shareholder, your position is optimally aligned with institutional projections. Given the 100% confidence rating in our BUY verdict and the significant +25.8% upside projected by the Wall Street Consensus, Veridian advises accumulating additional shares on any minor pullbacks. The objective is to maximize exposure ahead of the projected move toward the 2356.73 target.



Company or Ticker...

Hold? (y/n)

PREDICT ☒



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AI Chat



Profile



Analysts rating for Bharti Airtel derived from IND Money:

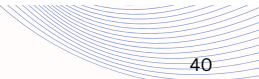
Analyst Rating for Bharti Airtel Ltd stock

based on 29 analysts



Based on 29 analysts offering long term price targets for Bharti Airtel Ltd. An average target of ₹2375

Source: Reuters LSEG Refinitiv



DASHBOARD

VERIDIAN 41

Hello, Navaneeth

Dashboard

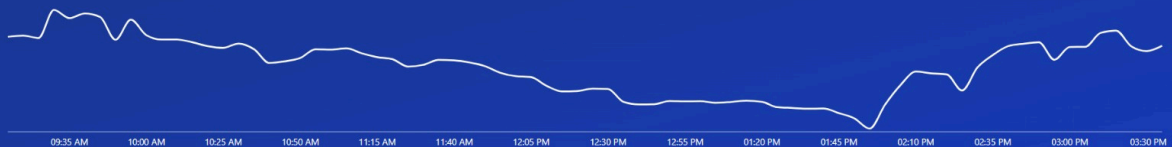
Overall Market My Portfolio

Q Search stocks (e.g. RELIANCE)...

NIFTY 50

₹24,865.7

₹312.95(-1.24%) Today



Home

AI Chat

Profile

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Top Gainers		Top Losers
LT	LT L&T	₹4,066.7 -4.95%
RE	RELIANCE Reliance	₹1,358 -2.58%
SB	SBIN SBI	₹1,189.9 -0.98%
HD	HDFCBANK HDFC Bank	₹879.4 -0.94%
TC	TCS TCS	₹2,613.5 -0.91%

Market News

Gold rallies towards \$5,300 as Mideast war Oil spike stoke haven bid

Home AI Chat Profile

Market News



Gold rallies towards \$5,300 as Mideast war, Oil spike stoke haven bid
FXStreet • 10:55 PM



Gold (XAUUSD), Silver, Platinum Forecasts - Silver Dives 7% As Traders Focus On Rising Energy Prices
FXEmpire • 10:35 PM

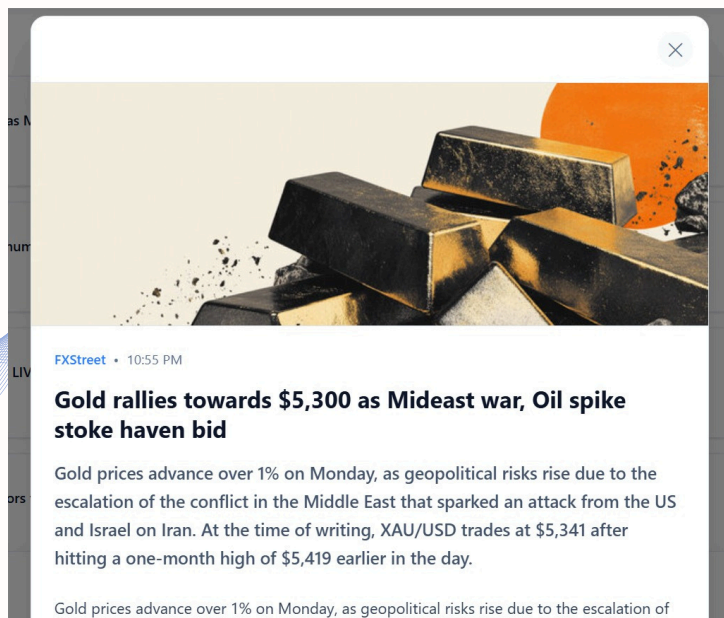


Iran-US war impact on India LIVE: IndiGo to operate 10 special flights from Jeddah to Indian cities on March 3
Hindustan Times • 09:55 PM



Wall Street slumps as investors fear prolonged Middle East conflict, energy & defense stocks rally
Mint • 08:11 PM





ASSUMPTIONS

- Financial data (price, fundamentals, sentiment, macro, alternative data) is available, reliable, and properly preprocessed.
- Each student model captures a distinct and meaningful market dimension.
- Model probabilities are calibrated, and sharper distributions imply higher confidence.
- Fixed reliability weights reasonably reflect long-term robustness of each analytical signal.
- Combining heterogeneous signals improves stability over single-model predictions.
- Market behavior is influenced by structural (fundamental/macro) and behavioral (sentiment/alternative) factors.
- Recommendations are decision-support signals, not guarantees of profit.

WORK BREAKDOWN

Vijai Kiren S

News and Sentiment
Analyzers + Market
Cycle + Expert
Consensus

Neha George

Macroeconomics
and Country risk+
Market Cycle +
Expert Consensus

Sarath Menon

Fundamentals and
Technical Analysis+
Market Cycle +
Expert Consensus

Navaneeth KB

Alternative and Web
Data model +
Teacher Model

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HARDWARE REQUIREMENT

- Laptop / Desktop
- Minimum 8 GB RAM (16 GB recommended)
- Intel i5 / Ryzen 5 or higher
- Minimum 10 GB free storage
- Stable Internet connection
- Google Colab (Cloud GPU support)

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SOFTWARE REQUIREMENT

◆ Programming Environment

- Python 3.x
- Google Colab
- VS Code

◆ ML & Data Libraries

- NumPy, Pandas
- Scikit-learn
- XGBoost
- PyTorch

◆ NLP & Explanation Module

- HuggingFace Transformers
- Google FLAN-T5 (Explanation Generation)

◆ Backend & Database

- FastAPI
- Firebase Authentication
- Firebase Firestore

◆ Data APIs

- Yahoo Finance API (Market & Fundamental Data)
- World Bank API (Macroeconomic Data)
- News APIs / RSS Feeds (Sentiment Data)

DATASET'S USED

Fundamental Data

Source: Yahoo Finance

Includes:

- Revenue growth
- ROA, ROE
- P/E ratio
- Debt-to-Equity ratio
- Free Cash Flow
- Profitability metrics

Used For:

- Financial health scoring
- Long-term intrinsic value analysis
- Risk assessment

Macroeconomic Data

Source: World Bank API

Includes:

- GDP growth rate
- Inflation rate
- Unemployment rate

Used For:

- Economic condition modeling
- Equity market outlook assessment
- Macro regime classification

Expert Consensus Data

Source: Yahoo Finance

Includes:

- Strong Buy
- Buy
- Hold
- Sell
- Strong Sell ratings

Used For:

- Analyst-weighted scoring
- Institutional sentiment modeling
- Confidence calibration

Alternative Data

Sources: Reddit, Google Trends, GitHub

Includes:

- Reddit sentiment intensity
- Search trend volume
- Developer activity (commits)

Used For:

- Public interest tracking
- Early signal detection
- Behavioral trend modeling

Sentiment Data

Source: News APIs / RSS Feeds

Includes:

- Financial news articles
- Online media reports
- Social media discussions

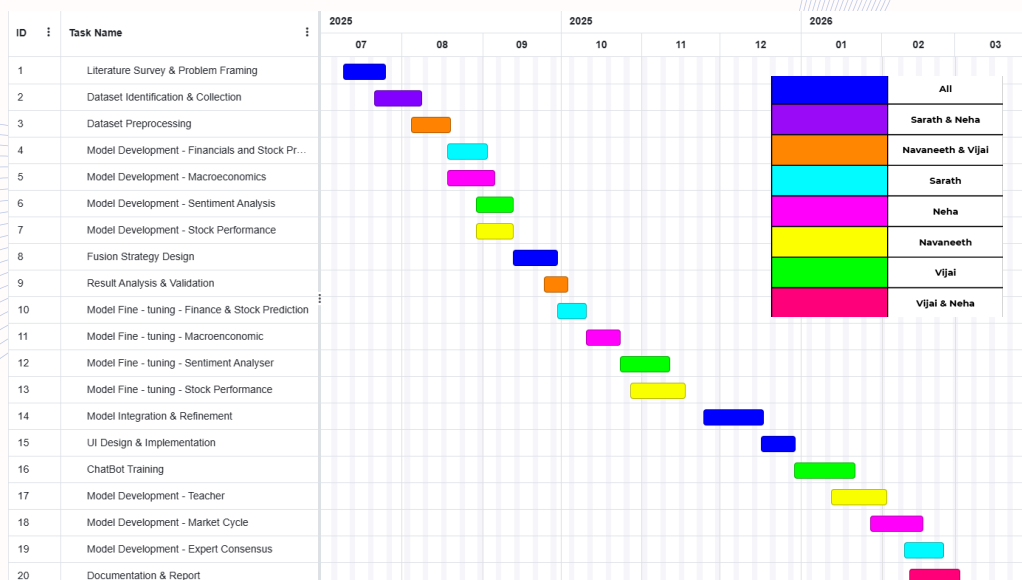
Used For:

- Sentiment classification (Positive / Neutral / Negative)
- Market mood analysis
- Improving prediction robustness

RISKS AND CHALLENGES

- **Data Accuracy Risk:** Reliance on third-party financial data sources; inaccurate data can mislead users.
- **AI Bias & Transparency:** Risk of biased or unclear recommendations if models are not explainable.
- **User Trust:** Convincing retail investors to trust AI-driven financial advice.
- **Security Concerns:** Protecting sensitive financial and personal data from breaches.
- **Regulatory & Compliance Issues:** Meeting financial regulations and data privacy laws (e.g., GDPR).
- **Market Volatility:** Sudden shifts in economic conditions may reduce prediction accuracy.
- **Scalability Challenges:** Handling large datasets and real-time processing as user base grows.
- **Integration Complexity:** Difficulty in combining multiple models

GANTT CHART



CONCLUSION

- Veridian presents an AI-powered approach to intelligent stock analysis and portfolio decision support.
- The system combines multiple financial perspectives into a single structured framework.
- A teacher-student ensemble model enhances stability and reduces conflicting signals.
- Confidence-weighted recommendations improve interpretability and transparency.
- The modular architecture allows scalability and future expansion.
- Veridian demonstrates how AI can assist investors in making more informed and structured decisions

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THANK YOU

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Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

- 1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles

of mathematics, natural sciences and engineering sciences

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project management and finance: Apply knowledge and understanding of en-

engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES:

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

**MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs)
AND PROGRAM SPECIFIC OUTCOMES (PSOs)**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3							3	2	2	1				3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2				1	3	3			
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

MAPPING	LEVEL	JUSTIFICATION
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.

101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.

101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.

101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.